

RTM V1 — Section 1

Document Overview & Traceability Scope

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

1

Version:

V1

Status:

APPROVED

1.1 Purpose

This document establishes the formal Requirements Traceability Matrix (RTM) framework for TODP.

The RTM provides end-to-end traceability across all project artifacts, ensuring that every business requirement can be traced through design, implementation, testing, governance validation, and production release.

The RTM serves as the authoritative traceability reference for:

- Product Management
- Engineering
- Quality Assurance
- Governance Teams
- Compliance Teams
- Security Teams
- Auditors
- Enterprise Customers

1.2 Document Objectives

RTM-OBJ-01

Provide complete requirement traceability.

RTM-OBJ-02

Provide implementation visibility.

RTM-OBJ-03

Provide verification traceability.

RTM-OBJ-04

Provide governance traceability.

RTM-OBJ-05

Provide auditability.

RTM-OBJ-06

Provide release readiness verification.

1.3 Traceability Principles

TODP traceability shall be based on the following principles:

Principle 1

Complete Coverage

Every approved requirement must be traceable.

Principle 2

Forward Traceability

Requirements must trace forward into implementation.

Principle 3

Backward Traceability

Implemented functionality must trace back to approved requirements.

Principle 4

Governance Traceability

Governance decisions must be traceable.

Principle 5

Auditability

All traceability records must be auditable.

Principle 6

Reproducibility

Traceability must support replay and verification.

1.4 RTM Scope

The RTM covers:

Business Requirements

System Requirements

Architecture Components

Application Components

Database Entities

API Contracts

Security Controls

Governance Controls

Audit Controls

Testing Artifacts

Release Artifacts

1.5 Traceability Domains

The RTM governs traceability across:

Domain 1

Business Domain

Domain 2

System Domain

Domain 3

Architecture Domain

Domain 4

Application Domain

Domain 5

Data Domain

Domain 6

API Domain

Domain 7

Testing Domain

Domain 8

Governance Domain

Domain 9

Release Domain

1.6 Authoritative Source Documents

The RTM shall trace requirements originating from:

PRD V1

Business Requirements Document (BRD V1)

Software Requirements Specification (SRS V1)

Software Architecture Document (SAD V1)

Technical Design Document (TDD V1)

Technical Reference Model (TRM V1)

Test Specification (TEST SPEC V1)

1.7 Traceability Hierarchy

TODP shall maintain the following hierarchy:

Business Objective



Business Requirement



System Requirement



Architecture Component



Technical Component



Implementation Artifact



Test Artifact



Governance Validation



Release Validation

1.8 RTM Coverage Model

Coverage shall include:

Requirement Coverage

Design Coverage

Implementation Coverage

Testing Coverage

Governance Coverage

Release Coverage

Audit Coverage

1.9 Forward Traceability Model

Forward traceability shall validate:

Requirement

↓

Architecture

↓

Design



Implementation



Testing



Deployment



Release

1.10 Backward Traceability Model

Backward traceability shall validate:

Release

↓

Deployment

↓

Testing

↓

Implementation

↓

Design

↓

Architecture

↓

Requirement

1.11 Traceability Record Structure

Each traceability record shall contain:

Trace ID

Requirement ID

Source Artifact

Target Artifact

Relationship Type

Status

Owner

Version

Approval Status

Audit Metadata

1.12 Relationship Types

Supported relationships include:

Defines

Implements

Depends On

Verifies

Approves

Governs

References

Supersedes

Extends

1.13 Traceability Lifecycle

Requirement Created

↓

Requirement Approved



Requirement Implemented



Requirement Verified



Requirement Governed



Requirement Released



Requirement Archived

1.14 Traceability Ownership

Ownership shall be assigned to:

Product Team

Engineering Team

QA Team

Governance Team

Security Team

Release Team

1.15 Traceability Validation Rules

RTM-RULE-01

Every requirement must have a unique identifier.

RTM-RULE-02

Every requirement must map to at least one implementation artifact.

RTM-RULE-03

Every implementation artifact must map to at least one requirement.

RTM-RULE-04

Every requirement must map to verification evidence.

RTM-RULE-05

Every requirement must map to governance evidence.

RTM-RULE-06

Every released capability must be traceable.

1.16 Traceability Metrics

The RTM shall measure:

Requirement Coverage %

Implementation Coverage %

Test Coverage %

Governance Coverage %

Release Coverage %

Audit Coverage %

1.17 Audit Requirements

The RTM must support:

Audit Reviews

Compliance Reviews

Governance Reviews

Security Reviews

Enterprise Due Diligence

Release Reviews

1.18 Compliance Requirements

The RTM shall support:

SOC2 Readiness

ISO 27001 Readiness

Enterprise Procurement Reviews

Customer Security Reviews

Internal Governance Reviews

1.19 RTM Success Criteria

The RTM is considered successful when:

100% Requirements Traced

100% Implementations Traced

100% Tests Traced

100% Releases Traced

100% Governance Controls Traced

100% Audit Evidence Available

1.20 Governance Rules

RTM-GOV-01

Traceability is mandatory.

RTM-GOV-02

Untraceable requirements are prohibited.

RTM-GOV-03

Unverified requirements are prohibited.

RTM-GOV-04

Unapproved releases are prohibited.

RTM-GOV-05

Audit evidence must be retained.

RTM-GOV-06

Traceability records must be reproducible.

1.21 Architectural Constraints

AC-RTM-01

Every requirement must have an identifier.

AC-RTM-02

Every artifact must support traceability.

AC-RTM-03

Traceability gaps require remediation.

AC-RTM-04

Governance controls cannot be bypassed.

AC-RTM-05

Audit records cannot be deleted.

AC-RTM-06

Release approval requires traceability validation.

1.22 Section Summary

This section defines the purpose, scope, objectives, principles, governance rules, lifecycle, ownership model, compliance requirements, and foundational traceability framework for RTM V1.

The RTM establishes the authoritative mechanism for tracing requirements from business intent through architecture, implementation, testing, governance validation, audit verification, and production release.

RTM V1 Progress

▣ Section 1 — Document Overview & Traceability Scope

▣ Section 2 — Traceability Framework

RTM V1 — Section 2

Traceability Framework

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

2

Version:

V1

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APPROVED

2.1 Purpose

This section defines the enterprise traceability framework used by TODP.

The framework establishes how requirements, designs, implementations, tests, governance controls, audits, and releases are connected through a structured traceability model.

The objective is to ensure:

- Complete visibility
- Controlled change management
- Verification coverage
- Governance compliance
- Audit readiness
- Release confidence

2.2 Framework Objectives

RTM-FRAMEWORK-01

Establish end-to-end traceability.

RTM-FRAMEWORK-02

Provide artifact relationship visibility.

RTM-FRAMEWORK-03

Support governance validation.

RTM-FRAMEWORK-04

Support auditability.

RTM-FRAMEWORK-05

Support impact analysis.

RTM-FRAMEWORK-06

Support release validation.

2.3 Enterprise Traceability Architecture

The TODP traceability architecture shall be:

Business Driven



Requirement Driven



Architecture Driven



Implementation Driven



Verification Driven



Governance Driven



Release Driven

2.4 Master Traceability Chain

The authoritative traceability chain shall be:

Business Objective



Business Requirement



System Requirement



Architecture Component



Technical Component



Implementation Artifact



Database Entity



API Contract



Test Case



Execution Evidence



Governance Approval



Release Approval

2.5 Traceability Layers

Layer 1

Business Traceability

Layer 2

Requirement Traceability

Layer 3

Architecture Traceability

Layer 4

Design Traceability

Layer 5

Implementation Traceability

Layer 6

Testing Traceability

Layer 7

Governance Traceability

Layer 8

Release Traceability

2.6 Artifact Classification Model

Artifact Categories:

Business Artifacts

Requirement Artifacts

Architecture Artifacts

Technical Artifacts

Implementation Artifacts

Testing Artifacts

Governance Artifacts

Release Artifacts

Audit Artifacts

2.7 Traceability Direction Types

Supported directions:

Forward Traceability

Backward Traceability

Bidirectional Traceability

Cross-Domain Traceability

Governance Traceability

Audit Traceability

2.8 Forward Traceability Framework

Forward traceability shall validate:

Requirement



Design



Implementation



Verification



Release

Purpose:

Ensure every requirement is implemented and verified.

2.9 Backward Traceability Framework

Backward traceability shall validate:

Release

↓

Implementation

↓

Design

↓

Requirement

Purpose:

Ensure every delivered capability has approved origin.

2.10 Bidirectional Traceability Framework

Bidirectional traceability shall support:

Requirement ↔ Architecture

Requirement ↔ Implementation

Requirement ↔ Test Cases

Requirement ↔ Governance Controls

Requirement ↔ Release Records

2.11 Cross-Domain Traceability

Cross-domain relationships shall support:

Business ↔ Technology

Architecture ↔ Security

Architecture ↔ Governance

Governance ↔ Testing

Release ↔ Audit

2.12 Traceability Identifier Framework

Every traceable artifact shall contain:

Unique ID

Artifact Type

Version

Owner

Status

Approval State

Audit Metadata

2.13 Traceability Identifier Standards

Examples:

BRD-001

SRS-001

SAD-001

TDD-001

API-001

DB-001

TC-001

REL-001

AUD-001

Identifiers must remain immutable.

2.14 Traceability Mapping Rules

MAP-01

One requirement may map to multiple artifacts.

MAP-02

One artifact may support multiple requirements.

MAP-03

Mappings must be explicitly documented.

MAP-04

Mappings must be auditable.

MAP-05

Mappings must be version controlled.

2.15 Traceability Relationship Model

Supported relationships:

Defines

Implements

Verifies

Approves

Depends On

Extends

Supersedes

Governs

Audits

Releases

2.16 Traceability Status Model

Statuses:

Draft

Approved

Implemented

Verified

Governed

Released

Archived

2.17 Version Traceability Framework

Every artifact version shall maintain:

Previous Version

Current Version

Approval State

Change History

Traceability History

2.18 Change Impact Framework

Changes shall support impact analysis across:

Requirements

Architecture

Components

APIs

Database Entities

Tests

Governance Controls

Releases

2.19 Impact Analysis Flow

Requirement Change



Architecture Impact



Implementation Impact



Testing Impact



Governance Impact



Release Impact

2.20 Coverage Validation Framework

Coverage shall be measured across:

Requirements

Architecture

Implementation

Testing

Governance

Release

Audit

2.21 Requirement Coverage Model

Requirement Coverage Formula:

Covered Requirements

÷

Total Requirements

×

100

2.22 Implementation Coverage Model

Implementation Coverage Formula:

Implemented Requirements

÷

Approved Requirements

×

100

2.23 Verification Coverage Model

Verification Coverage Formula:

Verified Requirements

÷

Implemented Requirements

×

100

2.24 Governance Coverage Model

Governance Coverage Formula:

Governed Requirements

÷

Verified Requirements

×

100

2.25 Release Coverage Model

Release Coverage Formula:

Released Requirements

÷

Approved Requirements

×

100

2.26 Audit Traceability Framework

Audit validation shall support:

Requirement Audits

Architecture Audits

Implementation Audits

Test Audits

Governance Audits

Release Audits

2.27 Compliance Traceability Framework

Traceability shall support:

SOC2 Controls

ISO27001 Controls

Enterprise Security Reviews

Customer Audits

Internal Governance Reviews

2.28 Release Validation Framework

Release approval requires:

Requirement Coverage

Implementation Coverage

Verification Coverage

Governance Coverage

Audit Coverage

2.29 Traceability Repository Requirements

The RTM repository shall support:

Search

Filtering

Version History

Relationship Navigation

Impact Analysis

Audit Reviews

2.30 Reporting Requirements

The framework shall provide:

Coverage Reports

Gap Reports

Impact Reports

Governance Reports

Release Reports

Audit Reports

2.31 Quality Validation Rules

RTM-QA-01

All mappings must be valid.

RTM-QA-02

All mappings must be current.

RTM-QA-03

All mappings must be auditable.

RTM-QA-04

All mappings must support replay.

RTM-QA-05

All mappings must support governance review.

2.32 Governance Rules

RTM-FW-GOV-01

Every artifact requires traceability.

RTM-FW-GOV-02

Traceability gaps are prohibited.

RTM-FW-GOV-03

Version history must be retained.

RTM-FW-GOV-04

Impact analysis is mandatory.

RTM-FW-GOV-05

Coverage reporting is mandatory.

RTM-FW-GOV-06

Audit evidence must be retained.

2.33 Architectural Constraints

AC-RTM-FW-01

Unmapped artifacts are prohibited.

AC-RTM-FW-02

Traceability identifiers cannot change.

AC-RTM-FW-03

Coverage gaps require remediation.

AC-RTM-FW-04

Governance validation cannot be bypassed.

AC-RTM-FW-05

Audit records cannot be deleted.

AC-RTM-FW-06

Release approval requires full traceability validation.

2.34 Section Summary

This section defines the enterprise traceability framework, mapping model, relationship model, identifier standards, coverage model, impact analysis framework, governance controls, audit requirements, and release validation mechanisms used throughout TODP.

The Traceability Framework serves as the foundational structure that connects all business, technical, governance, testing, audit, and release artifacts into a single auditable and reproducible system of record.

RTM V1 Progress

▣ Section 1 — Document Overview & Traceability Scope

▣ Section 2 — Traceability Framework

▣ Section 3 — Requirement Classification Model

RTM V1 — Section 3

Requirement Classification Model

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

3

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APPROVED

3.1 Purpose

This section defines the Requirement Classification Model used throughout TODP.

Requirement classification establishes a structured approach for organizing, managing, tracing, governing, verifying, auditing, and releasing requirements.

The objective is to ensure:

- Consistent requirement categorization
- Traceability consistency
- Governance consistency
- Coverage measurement
- Audit readiness
- Release readiness

3.2 Classification Objectives

RTM-CLASS-01

Provide requirement structure.

RTM-CLASS-02

Support traceability.

RTM-CLASS-03

Support impact analysis.

RTM-CLASS-04

Support governance controls.

RTM-CLASS-05

Support verification planning.

RTM-CLASS-06

Support audit reviews.

3.3 Requirement Hierarchy

TODP requirements shall be organized as:

Business Objective



Business Requirement



Capability Requirement



System Requirement



Technical Requirement



Implementation Requirement



Verification Requirement



Governance Requirement

3.4 Requirement Domains

Requirements shall be classified into:

Business Domain

Functional Domain

Technical Domain

Security Domain

Governance Domain

Operational Domain

Compliance Domain

Quality Domain

3.5 Business Requirements

Identifier Prefix:

BRD

Purpose:

Define business goals, stakeholder needs, and organizational objectives.

Examples:

Opportunity Management

Decision Transparency

Recommendation Explainability

Governance Compliance

Auditability

3.6 Functional Requirements

Identifier Prefix:

SRS-FR

Purpose:

Define platform behaviors and user-visible capabilities.

Examples:

Create Opportunity

Search Evidence

Execute Ranking

Generate Recommendation

Replay Decision

3.7 Non-Functional Requirements

Identifier Prefix:

SRS-NFR

Purpose:

Define quality attributes and operational expectations.

Examples:

Performance

Scalability

Reliability

Availability

Maintainability

3.8 Technical Requirements

Identifier Prefix:

TDD

Purpose:

Define implementation-level technical behaviors.

Examples:

JWT Authentication

Hybrid Retrieval

PostgreSQL Persistence

Redis Caching

Audit Storage

3.9 Architecture Requirements

Identifier Prefix:

SAD

Purpose:

Define architectural structures and component interactions.

Examples:

Layered Architecture

Service Boundaries

Domain Separation

Data Flow Controls

Integration Controls

3.10 Security Requirements

Identifier Prefix:

SEC

Purpose:

Define protection mechanisms and security controls.

Examples:

Authentication

Authorization

Encryption

Secrets Management

Audit Protection

3.11 Governance Requirements

Identifier Prefix:

GOV

Purpose:

Define policy, approval, compliance, and oversight controls.

Examples:

Approval Requirements

Compliance Controls

Exception Handling

Governance Reporting

Policy Enforcement

3.12 Audit Requirements

Identifier Prefix:

AUD

Purpose:

Define auditability and traceability controls.

Examples:

Audit Logs

Snapshots

Replay Support

Traceability Records

Evidence Preservation

3.13 Operational Requirements

Identifier Prefix:

OPS

Purpose:

Define production operational expectations.

Examples:

Monitoring

Alerting

Incident Management

Recovery

Deployment

3.14 Compliance Requirements

Identifier Prefix:

COMP

Purpose:

Define regulatory and compliance obligations.

Examples:

SOC2 Readiness

ISO27001 Readiness

Retention Controls

Access Controls

Evidence Controls

3.15 Test Requirements

Identifier Prefix:

TEST

Purpose:

Define verification obligations.

Examples:

Unit Testing

Integration Testing

Security Testing

Performance Testing

UAT Testing

3.16 Release Requirements

Identifier Prefix:

REL

Purpose:

Define deployment and release obligations.

Examples:

Approval Gates

Coverage Validation

Release Evidence

Rollback Validation

Production Readiness

3.17 Requirement Criticality Levels

Supported Levels:

Critical

High

Medium

Low

Informational

3.18 Critical Requirement Definition

Critical requirements:

- Affect security
- Affect governance
- Affect auditability
- Affect legal obligations
- Affect business continuity

Critical requirements require:

Mandatory Testing

Mandatory Governance Review

Mandatory Audit Review

Mandatory Release Approval

3.19 Priority Levels

Supported priorities:

P1

P2

P3

P4

P5

Definitions:

P1 = Highest Priority

P5 = Lowest Priority

3.20 Requirement Status Model

Statuses:

Draft

Review

Approved

Implemented

Verified

Governed

Released

Archived

3.21 Requirement Ownership Model

Every requirement shall have:

Business Owner

Technical Owner

Governance Owner

Verification Owner

Release Owner

3.22 Requirement Source Model

Requirements may originate from:

PRD

BRD

Stakeholder Reviews

Governance Reviews

Security Reviews

Audit Reviews

Operational Reviews

Compliance Reviews

3.23 Requirement Dependency Model

Dependencies shall be classified as:

Business Dependency

Technical Dependency

Security Dependency

Governance Dependency

Operational Dependency

Compliance Dependency

3.24 Requirement Mapping Model

Every requirement shall map to:

Architecture Component

Technical Component

Implementation Artifact

Test Artifact

Governance Control

Release Artifact

3.25 Requirement Verification Model

Verification methods include:

Inspection

Analysis

Demonstration

Testing

Governance Validation

Audit Validation

3.26 Requirement Traceability Model

Every requirement shall support:

Forward Traceability

Backward Traceability

Cross-Domain Traceability

Governance Traceability

Audit Traceability

3.27 Requirement Versioning Model

Requirements shall maintain:

Requirement Version

Change History

Approval History

Ownership History

Traceability History

3.28 Requirement Change Classification

Changes shall be classified as:

Minor

Moderate

Major

Critical

Impact analysis is mandatory for:

Major Changes

Critical Changes

3.29 Requirement Coverage Categories

Coverage shall be measured for:

Requirement Coverage

Implementation Coverage

Verification Coverage

Governance Coverage

Release Coverage

Audit Coverage

3.30 Requirement Repository Structure

The RTM repository shall organize requirements by:

Domain

Type

Priority

Criticality

Status

Owner

Version

3.31 Governance Rules

RTM-CLASS-GOV-01

Every requirement must be classified.

RTM-CLASS-GOV-02

Every requirement must have ownership.

RTM-CLASS-GOV-03

Every requirement must have traceability.

RTM-CLASS-GOV-04

Critical requirements require enhanced governance.

RTM-CLASS-GOV-05

Changes require impact analysis.

RTM-CLASS-GOV-06

Coverage must be measurable.

3.32 Architectural Constraints

AC-RTM-CLASS-01

Unclassified requirements are prohibited.

AC-RTM-CLASS-02

Requirements without ownership are prohibited.

AC-RTM-CLASS-03

Requirements without verification are prohibited.

AC-RTM-CLASS-04

Critical requirements cannot bypass governance.

AC-RTM-CLASS-05

Version history must be retained.

AC-RTM-CLASS-06

Release approval requires requirement validation.

3.33 Section Summary

This section defines the Requirement Classification Model, including requirement types, domains, ownership structures, criticality levels, priority models, dependency structures, verification models, governance controls, and traceability obligations.

The classification framework provides the foundation for consistent requirement management, impact analysis, auditability, governance enforcement, verification planning, and release readiness throughout TODP.

RTM V1 Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Framework

☐ Section 3 — Requirement Classification Model

☐ Section 4 — BRD Traceability Matrix

RTM V1 — Section 4

BRD Traceability Matrix

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

4

Version:

V1

Status:

APPROVED

4.1 Purpose

This section establishes traceability between Business Requirements defined in BRD V1 and all downstream implementation artifacts.

The objective is to ensure that every approved business requirement:

- Is implemented
- Is verified
- Is governed
- Is auditable
- Is releasable
- Is traceable end-to-end

4.2 Business Requirement Traceability Model

Business Objective



Business Requirement



System Requirement



Architecture Component



Technical Design



Implementation



Testing



Governance Validation



Release Approval

4.3 BRD Traceability Domains

The following domains shall be traced:

Business Goals

Stakeholder Objectives

Business Capabilities

Operational Requirements

Governance Requirements

Compliance Requirements

Audit Requirements

Strategic Requirements

4.4 Master BRD Traceability Structure

Each BRD requirement shall map to:

BRD Requirement ID

↓

SRS Requirement ID

↓

SAD Component ID

↓

TDD Component ID

↓

API ID



Database Entity



Test Case



Governance Control



Release Record

4.5 BRD Requirement Categories

Category BRD-01

Opportunity Management

Category BRD-02

Authority Management

Category BRD-03

Evidence Management

Category BRD-04

Retrieval Management

Category BRD-05

Ranking Management

Category BRD-06

Recommendation Management

Category BRD-07

Governance Management

Category BRD-08

Audit & Replay Management

4.6 Opportunity Requirement Traceability Matrix

BRD-OPP-001

Manage Opportunities

↓

SRS-FR-OPP-001

Opportunity CRUD



SAD-COMP-OPP

Opportunity Service



TDD-OPP

Opportunity Design



API-OPP-001

Opportunity API



DB-OPPORTUNITY

Opportunity Table



TEST-OPP-001

Opportunity Test Suite



GOV-OPP-001

Opportunity Governance



REL-OPP-001

Release Validation

4.7 Authority Requirement Traceability Matrix

BRD-AUTH-001

Manage Authorities



SRS-FR-AUTH-001

Authority CRUD



SAD-COMP-AUTH

Authority Service



TDD-AUTH

Authority Design



API-AUTH-001

Authority API



DB-AUTHORITY

Authority Table



TEST-AUTH-001

Authority Test Suite



GOV-AUTH-001

Authority Governance



REL-AUTH-001

Release Validation

4.8 Evidence Requirement Traceability Matrix

BRD-EVD-001

Manage Evidence



SRS-FR-EVD-001

Evidence CRUD



SAD-COMP-EVD

Evidence Service



TDD-EVD

Evidence Design



API-EVD-001

Evidence API



DB-EVIDENCE

Evidence Table



TEST-EVD-001

Evidence Test Suite



GOV-EVD-001

Evidence Governance



REL-EVD-001

Release Validation

4.9 Retrieval Requirement Traceability Matrix

BRD-RET-001

Search Information



SRS-FR-RET-001

Hybrid Retrieval



SAD-COMP-RET

Retrieval Engine



TDD-RET

Retrieval Design



API-RET-001

Retrieval API



DB-EMBEDDINGS

Embedding Store



TEST-RET-001

Retrieval Tests



GOV-RET-001

Retrieval Governance



REL-RET-001

Release Validation

4.10 Ranking Requirement Traceability Matrix

BRD-RANK-001

Rank Opportunities



SRS-FR-RANK-001

Ranking Execution



SAD-COMP-RANK

Ranking Engine



TDD-RANK

Ranking Design



API-RANK-001

Ranking API



DB-RANKING

Ranking Storage



TEST-RANK-001

Ranking Tests



GOV-RANK-001

Ranking Governance



REL-RANK-001

Release Validation

BRD-REC-001

Generate Recommendations



SRS-FR-REC-001

Recommendation Execution



SAD-COMP-REC

Recommendation Engine



TDD-REC

Recommendation Design



API-REC-001

Recommendation API



DB-RECOMMENDATION

Recommendation Storage



TEST-REC-001

Recommendation Tests



GOV-REC-001

Recommendation Governance



REL-REC-001

Release Validation

4.12 Governance Requirement Traceability Matrix

BRD-GOV-001

Governance Decisions



SRS-FR-GOV-001

Governance Workflows



SAD-COMP-GOV

Governance Service



TDD-GOV

Governance Design



API-GOV-001

Governance API



DB-GOVERNANCE

Governance Storage



TEST-GOV-001

Governance Tests



GOV-GOV-001

Governance Controls



REL-GOV-001

Release Validation

4.13 Audit Requirement Traceability Matrix

BRD-AUD-001

Audit Decisions



SRS-FR-AUD-001

Audit Logging



SAD-COMP-AUD

Audit Service



TDD-AUD

Audit Design



API-AUD-001

Audit API



DB-AUDIT

Audit Storage



TEST-AUD-001

Audit Tests



GOV-AUD-001

Audit Governance



REL-AUD-001

Release Validation

4.14 Replay Requirement Traceability Matrix

BRD-REP-001

Replay Decisions

↓

SRS-FR-REP-001

Replay Execution

↓

SAD-COMP-REP

Replay Engine

↓

TDD-REP

Replay Design



API-REP-001

Replay API



DB-SNAPSHOT

Snapshot Storage



TEST-REP-001

Replay Tests



GOV-REP-001

Replay Governance



REL-REP-001

Release Validation

4.15 Security Requirement Traceability Matrix

BRD-SEC-001

Secure Platform

↓

SRS-NFR-SEC-001

Security Controls

↓

SAD-SECURITY

Security Architecture

↓

TDD-SEC

Security Design



API-AUTHN

Authentication APIs



DB-SECURITY

Security Data



TEST-SEC-001

Security Tests



GOV-SEC-001

Security Governance



REL-SEC-001

Release Validation

4.16 Compliance Requirement Traceability Matrix

BRD-COMP-001

Compliance Readiness



SRS-NFR-COMP-001

Compliance Controls



SAD-COMP

Compliance Architecture



TDD-COMP

Compliance Design



API-COMP

Compliance APIs



DB-COMP

Compliance Data



TEST-COMP-001

Compliance Tests



GOV-COMP-001

Compliance Governance



REL-COMP-001

Release Validation

4.17 Traceability Coverage Requirements

Every BRD requirement shall trace to:

Minimum 1 SRS Requirement

Minimum 1 Architecture Component

Minimum 1 Technical Component

Minimum 1 Test Artifact

Minimum 1 Governance Control

Minimum 1 Release Record

4.18 Coverage Validation Matrix

Coverage Type



Business Coverage



Functional Coverage



Architecture Coverage



Implementation Coverage



Testing Coverage



Governance Coverage



Release Coverage

4.19 Gap Detection Rules

A traceability gap exists when:

Requirement Missing Mapping

Implementation Missing Mapping

Test Missing Mapping

Governance Missing Mapping

Release Missing Mapping

All gaps require remediation.

4.20 Change Impact Rules

Changes to BRD requirements shall trigger review of:

SRS

SAD

TDD

API Specifications

Database Designs

Tests

Governance Controls

Release Plans

4.21 Coverage Metrics

Required Metrics:

Business Requirement Coverage %

Implementation Coverage %

Verification Coverage %

Governance Coverage %

Release Coverage %

Target:

100%

4.22 Audit Requirements

BRD traceability must support:

Business Audits

Governance Audits

Security Audits

Compliance Audits

Release Audits

Enterprise Due Diligence

4.23 Governance Rules

RTM-BRD-GOV-01

Every BRD requirement must be traced.

RTM-BRD-GOV-02

Every BRD requirement must be verified.

RTM-BRD-GOV-03

Every BRD requirement must have governance validation.

RTM-BRD-GOV-04

Every BRD requirement must have release validation.

RTM-BRD-GOV-05

Coverage must be measurable.

RTM-BRD-GOV-06

Audit evidence must be retained.

4.24 Architectural Constraints

AC-RTM-BRD-01

Unmapped BRD requirements are prohibited.

AC-RTM-BRD-02

Unverified BRD requirements are prohibited.

AC-RTM-BRD-03

Untested BRD requirements are prohibited.

AC-RTM-BRD-04

Governance validation cannot be skipped.

AC-RTM-BRD-05

Traceability evidence cannot be deleted.

AC-RTM-BRD-06

Release approval requires complete traceability.

4.25 Section Summary

This section defines the Business Requirements Traceability Matrix and establishes the mapping between BRD requirements and downstream artifacts including SRS requirements, architecture components, technical designs, APIs, database entities, tests, governance controls, and release approvals.

The BRD Traceability Matrix ensures that every business objective within TODP is fully implemented, verified, governed, auditable, and releasable.

RTM V1 Progress

📄 Section 1 — Document Overview & Traceability Scope

📄 Section 2 — Traceability Framework

📄 Section 3 — Requirement Classification Model

❓ Section 4 — BRD Traceability Matrix

❓ Section 5 — SRS Traceability Matrix

RTM V1 — Section 5

SRS Traceability Matrix

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

5

Version:

V1

Status:

APPROVED

5.1 Purpose

This section establishes traceability between Software Requirements

defined in SRS V1 and all downstream implementation, verification, governance, and release artifacts.

The objective is to ensure that every approved system requirement is:

- Architected
- Designed
- Implemented
- Tested
- Governed
- Released

5.2 SRS Traceability Model

System Requirement



Architecture Component



Technical Design



Application Component



API Contract



Database Entity



Test Specification



Governance Validation



Release Approval

5.3 SRS Requirement Domains

The following requirement domains shall be traced:

Functional Requirements

Non-Functional Requirements

Security Requirements

Governance Requirements

Audit Requirements

Operational Requirements

Compliance Requirements

Integration Requirements

5.4 Functional Requirement Traceability Structure

Each functional requirement shall map to:

SRS Requirement

↓

SAD Component



TDD Component



API



Database Entity



Test Case



Governance Control



Release Validation

5.5 Functional Requirement Categories

SRS-FR-01

Opportunity Requirements

SRS-FR-02

Authority Requirements

SRS-FR-03

Evidence Requirements

SRS-FR-04

Retrieval Requirements

SRS-FR-05

Ranking Requirements

SRS-FR-06

Recommendation Requirements

SRS-FR-07

Governance Requirements

SRS-FR-08

Audit & Replay Requirements

5.6 Opportunity Functional Traceability

SRS-FR-OPP-001

Opportunity CRUD



SAD-COMP-OPP

Opportunity Service



TDD-OPP

Opportunity Design



API-OPP

Opportunity APIs



DB-OPPORTUNITY

Opportunity Table



TEST-OPP

Opportunity Tests



GOV-OPP

Governance Validation



REL-OPP

Release Validation

5.7 Authority Functional Traceability

SRS-FR-AUTH-001

Authority Management



SAD-COMP-AUTH

Authority Service



TDD-AUTH

Authority Design



API-AUTH

Authority APIs



DB-AUTHORITY

Authority Table



TEST-AUTH

Authority Tests



GOV-AUTH

Governance Validation



REL-AUTH

Release Validation

5.8 Evidence Functional Traceability

SRS-FR-EVD-001

Evidence Management



SAD-COMP-EVD

Evidence Service



TDD-EVD

Evidence Design



API-EVD

Evidence APIs



DB-EVIDENCE

Evidence Table



TEST-EVD

Evidence Tests



GOV-EVD

Governance Validation



REL-EVD

Release Validation

5.9 Retrieval Functional Traceability

SRS-FR-RET-001

Hybrid Retrieval



SAD-COMP-RET

Retrieval Engine



TDD-RET

Retrieval Design



API-RET

Retrieval APIs



DB-EMBEDDING

Embedding Store



TEST-RET

Retrieval Tests



GOV-RET

Governance Validation



REL-RET

Release Validation

5.10 Ranking Functional Traceability

SRS-FR-RANK-001

Ranking Execution



SAD-COMP-RANK

Ranking Engine



TDD-RANK

Ranking Design



API-RANK

Ranking APIs



DB-RANKING

Ranking Store



TEST-RANK

Ranking Tests



GOV-RANK

Governance Validation



REL-RANK

Release Validation

5.11 Recommendation Functional Traceability

SRS-FR-REC-001

Recommendation Generation



SAD-COMP-REC

Recommendation Engine



TDD-REC

Recommendation Design



API-REC

Recommendation APIs



DB-RECOMMENDATION

Recommendation Store



TEST-REC

Recommendation Tests



GOV-REC

Governance Validation



REL-REC

Release Validation

5.12 Governance Functional Traceability

SRS-FR-GOV-001

Policy Enforcement



SAD-COMP-GOV

Governance Service



TDD-GOV

Governance Design



API-GOV

Governance APIs



DB-GOVERNANCE

Governance Store

↓

TEST-GOV

Governance Tests

↓

GOV-GOV

Governance Controls

↓

REL-GOV

Release Validation

5.13 Audit Functional Traceability

SRS-FR-AUD-001

Audit Logging



SAD-COMP-AUD

Audit Service



TDD-AUD

Audit Design



API-AUD

Audit APIs



DB-AUDIT

Audit Store



TEST-AUD

Audit Tests



GOV-AUD

Governance Validation



REL-AUD

Release Validation

5.14 Replay Functional Traceability

SRS-FR-REP-001

Replay Execution



SAD-COMP-REP

Replay Engine



TDD-REP

Replay Design



API-REP

Replay APIs



DB-SNAPSHOT

Snapshot Store



TEST-REP

Replay Tests



GOV-REP

Governance Validation



REL-REP

Release Validation

5.15 Non-Functional Requirement Traceability

Non-functional requirements shall map to:

Architecture Controls

Technical Controls

Infrastructure Controls

Security Controls

Operational Controls

Test Controls

Governance Controls

Release Controls

5.16 Performance Requirement Traceability

SRS-NFR-PERF-001

Performance Targets



SAD-PERFORMANCE

Performance Architecture



TDD-PERFORMANCE

Performance Design



INFRA-PERF

Infrastructure Controls



TEST-PERF

Performance Tests



REL-PERF

Release Validation

5.17 Scalability Requirement Traceability

SRS-NFR-SCALE-001

Scalability Targets



SAD-SCALABILITY

Scalability Architecture



TDD-SCALABILITY

Scalability Design



INFRA-SCALE

Infrastructure Controls



TEST-SCALE

Scalability Tests



REL-SCALE

Release Validation

5.18 Security Requirement Traceability

SRS-NFR-SEC-001

Security Controls

↓

SAD-SECURITY

Security Architecture

↓

TDD-SECURITY

Security Design

↓

API-AUTHN

Authentication APIs



DB-SECURITY

Security Data



TEST-SEC

Security Tests



GOV-SEC

Security Governance



REL-SEC

Release Validation

5.19 Availability Requirement Traceability

SRS-NFR-AVL-001

Availability Targets



SAD-HA

High Availability Architecture



TDD-HA

Availability Design



OPS-HA

Operational Controls



TEST-HA

Availability Tests



REL-HA

Release Validation

5.20 Auditability Requirement Traceability

SRS-NFR-AUD-001

Auditability Requirements



SAD-AUDIT

Audit Architecture



TDD-AUDIT

Audit Design



DB-AUDIT

Audit Storage



TEST-AUDIT

Audit Tests



REL-AUDIT

Release Validation

5.21 Traceability Coverage Requirements

Every SRS requirement shall map to:

Minimum 1 Architecture Component

Minimum 1 Design Component

Minimum 1 Implementation Artifact

Minimum 1 Verification Artifact

Minimum 1 Governance Artifact

Minimum 1 Release Artifact

5.22 Gap Detection Requirements

Traceability gaps exist when:

Architecture Missing

Design Missing

Implementation Missing

Testing Missing

Governance Missing

Release Missing

All gaps require remediation.

5.23 Impact Analysis Requirements

Changes to SRS requirements shall trigger review of:

Architecture

Design

Implementation

Database

APIs

Testing

Governance

Release Plans

5.24 Coverage Metrics

Required metrics:

Functional Coverage %

Non-Functional Coverage %

Verification Coverage %

Governance Coverage %

Release Coverage %

Audit Coverage %

Target:

100%

5.25 Verification Requirements

Each SRS requirement shall have:

Verification Method

Verification Owner

Verification Evidence

Verification Status

Verification History

5.26 Governance Requirements

Each SRS requirement shall have:

Governance Owner

Governance Validation

Approval Status

Compliance Status

Audit Status

5.27 Release Requirements

Each SRS requirement shall support:

Release Traceability

Release Validation

Release Approval

Release Auditability

Release History

5.28 Governance Rules

RTM-SRS-GOV-01

Every SRS requirement must be traced.

RTM-SRS-GOV-02

Every SRS requirement must be implemented.

RTM-SRS-GOV-03

Every SRS requirement must be tested.

RTM-SRS-GOV-04

Every SRS requirement must be governed.

RTM-SRS-GOV-05

Every SRS requirement must support release validation.

RTM-SRS-GOV-06

Coverage must be measurable.

5.29 Architectural Constraints

AC-RTM-SRS-01

Unmapped SRS requirements are prohibited.

AC-RTM-SRS-02

Untested SRS requirements are prohibited.

AC-RTM-SRS-03

Unverified SRS requirements are prohibited.

AC-RTM-SRS-04

Governance validation cannot be skipped.

AC-RTM-SRS-05

Audit evidence cannot be deleted.

AC-RTM-SRS-06

Release approval requires complete traceability.

5.30 Section Summary

This section defines the Software Requirements Traceability Matrix and establishes mappings between system requirements and architecture components, technical designs, APIs, database entities, verification artifacts, governance controls, and release approvals.

The SRS Traceability Matrix ensures that every system requirement within TODP is fully implemented, verified, governed, auditable, and release-ready.

RTM V1 Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Framework

☐ Section 3 — Requirement Classification Model

☐ Section 4 — BRD Traceability Matrix

☐ Section 5 — SRS Traceability Matrix

☐ Section 6 — Architecture Traceability Matrix

RTM V1 — Section 6

Architecture Traceability Matrix

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

Version:

V1

Status:

APPROVED

6.1 Purpose

This section defines the Architecture Traceability Matrix for TODP.

The Architecture Traceability Matrix establishes relationships between approved system requirements and architectural components defined in SAD V1.

The objective is to ensure that:

- Every architectural component has a business purpose
- Every architectural component satisfies approved requirements
- Every architectural component is testable
- Every architectural component is governable
- Every architectural component is auditable
- Every architectural component supports release readiness

6.2 Architecture Traceability Objectives

RTM-ARCH-01

Verify architecture coverage.

RTM-ARCH-02

Verify requirement realization.

RTM-ARCH-03

Verify component accountability.

RTM-ARCH-04

Verify governance alignment.

RTM-ARCH-05

Verify auditability.

RTM-ARCH-06

Verify release readiness.

6.3 Architecture Traceability Model

Business Requirement



System Requirement



Architecture Component



Technical Component



Implementation Artifact



Verification Artifact



Governance Artifact



Release Artifact

6.4 Architecture Domains

The following architectural domains shall be traced:

Presentation Layer

API Layer

Application Layer

Domain Layer

Persistence Layer

Search Layer

Governance Layer

Audit Layer

Infrastructure Layer

Security Layer

6.5 Presentation Layer Traceability

Architecture Component:

SAD-PRES-001

Frontend Presentation Layer

Maps To:

BRD-UI-001

User Experience Requirements

↓

SRS-FR-UI-001

User Interface Requirements



TDD-UI

Frontend Design



TEST-UI

UI Verification



REL-UI

Release Validation

6.6 API Layer Traceability

Architecture Component:

SAD-API-001

API Gateway Layer

Maps To:

SRS-FR-API-001

API Requirements

↓

TDD-API

API Design

↓

API-SPEC

API Contracts

↓

TEST-API

API Testing



REL-API

Release Validation

6.7 Authentication Architecture Traceability

Architecture Component:

SAD-AUTH-001

Authentication Service

Maps To:

SRS-NFR-SEC-001

Authentication Requirements



TDD-AUTH

Authentication Design



API-AUTHN

Authentication APIs



TEST-AUTH

Authentication Tests



GOV-SEC

Security Governance

6.8 Authorization Architecture Traceability

Architecture Component:

SAD-RBAC-001

Authorization Service

Maps To:

SRS-NFR-SEC-002

Authorization Requirements

↓

TDD-RBAC

RBAC Design

↓

API-RBAC

RBAC APIs

↓

TEST-RBAC

RBAC Tests



GOV-RBAC

RBAC Governance

6.9 Opportunity Service Traceability

Architecture Component:

SAD-COMP-OPP

Opportunity Service

Maps To:

BRD-OPP-001

Opportunity Management



SRS-FR-OPP-001

Opportunity Requirements



TDD-OPP

Opportunity Design



TEST-OPP

Opportunity Tests



REL-OPP

Release Validation

6.10 Authority Service Traceability

Architecture Component:

SAD-COMP-AUTH

Authority Service

Maps To:

BRD-AUTH-001

Authority Management

↓

SRS-FR-AUTH-001

Authority Requirements

↓

TDD-AUTH

Authority Design



TEST-AUTH

Authority Tests



REL-AUTH

Release Validation

6.11 Evidence Service Traceability

Architecture Component:

SAD-COMP-EVD

Evidence Service

Maps To:

BRD-EVD-001

Evidence Management



SRS-FR-EVD-001

Evidence Requirements



TDD-EVD

Evidence Design



TEST-EVD

Evidence Tests



REL-EVD

Release Validation

6.12 Retrieval Architecture Traceability

Architecture Component:

SAD-COMP-RET

Retrieval Engine

Maps To:

BRD-RET-001

Search Requirements

↓

SRS-FR-RET-001

Hybrid Retrieval

↓

TDD-RET

Retrieval Design



TEST-RET

Retrieval Tests



REL-RET

Release Validation

6.13 Ranking Architecture Traceability

Architecture Component:

SAD-COMP-RANK

Ranking Engine

Maps To:

BRD-RANK-001

Ranking Requirements

↓

SRS-FR-RANK-001

Ranking Execution

↓

TDD-RANK

Ranking Design

↓

TEST-RANK

Ranking Tests

↓

REL-RANK

Release Validation

6.14 Recommendation Architecture Traceability

Architecture Component:

SAD-COMP-REC

Recommendation Engine

Maps To:

BRD-REC-001

Recommendation Requirements

↓

SRS-FR-REC-001

Recommendation Generation



TDD-REC

Recommendation Design



TEST-REC

Recommendation Tests



REL-REC

Release Validation

6.15 Governance Architecture Traceability

Architecture Component:

SAD-COMP-GOV

Governance Service

Maps To:

BRD-GOV-001

Governance Requirements

↓

SRS-FR-GOV-001

Policy Enforcement

↓

TDD-GOV

Governance Design

↓

TEST-GOV

Governance Tests



REL-GOV

Release Validation

6.16 Audit Architecture Traceability

Architecture Component:

SAD-COMP-AUD

Audit Service

Maps To:

BRD-AUD-001

Audit Requirements



SRS-FR-AUD-001

Audit Logging



TDD-AUD

Audit Design



TEST-AUD

Audit Tests



REL-AUD

Release Validation

6.17 Replay Architecture Traceability

Architecture Component:

SAD-COMP-REP

Replay Engine

Maps To:

BRD-REP-001

Replay Requirements

↓

SRS-FR-REP-001

Replay Execution

↓

TDD-REP

Replay Design

↓

TEST-REP

Replay Tests



REL-REP

Release Validation

6.18 Search Infrastructure Traceability

Architecture Component:

SAD-INFRA-SEARCH

Hybrid Search Infrastructure

Maps To:

SRS-FR-RET-001

Hybrid Search



TDD-SEARCH

Search Design



DB-EMBEDDINGS

Vector Storage



TEST-SEARCH

Search Tests



REL-SEARCH

Release Validation

6.19 Database Architecture Traceability

Architecture Component:

SAD-INFRA-DB

PostgreSQL Layer

Maps To:

Persistence Requirements

↓

TDD-DATABASE

Database Design

↓

DB-SCHEMA

Database Entities



TEST-DB

Database Tests



REL-DB

Release Validation

6.20 Cache Architecture Traceability

Architecture Component:

SAD-INFRA-CACHE

Redis Layer

Maps To:

Performance Requirements



TDD-CACHE

Cache Design



CACHE-IMPLEMENTATION

Cache Layer



TEST-CACHE

Cache Tests



REL-CACHE

Release Validation

6.21 Observability Architecture Traceability

Architecture Component:

SAD-OBS-001

Observability Platform

Maps To:

Operational Requirements

↓

TDD-OBS

Observability Design

↓

LOGGING

Logging Components

↓

TEST-OBS

Observability Tests



REL-OBS

Release Validation

6.22 Security Architecture Traceability

Architecture Component:

SAD-SEC-001

Security Architecture

Maps To:

Security Requirements



TDD-SEC

Security Design



AUTHN/RBAC

Security Components



TEST-SEC

Security Tests



REL-SEC

Release Validation

6.23 Architecture Coverage Requirements

Every architecture component shall map to:

Minimum 1 Business Requirement

Minimum 1 System Requirement

Minimum 1 Technical Design

Minimum 1 Verification Artifact

Minimum 1 Governance Artifact

Minimum 1 Release Artifact

6.24 Architecture Gap Analysis

Architecture gaps exist when:

Requirement Not Architected

Component Without Requirement

Missing Verification

Missing Governance Validation

Missing Release Validation

All gaps require remediation.

6.25 Architecture Impact Analysis

Changes to architecture shall trigger review of:

Requirements

Technical Designs

Database Models

API Specifications

Test Specifications

Governance Controls

Release Plans

6.26 Architecture Coverage Metrics

Required Metrics:

Architecture Coverage %

Verification Coverage %

Governance Coverage %

Release Coverage %

Audit Coverage %

Target:

100%

6.27 Governance Rules

RTM-ARCH-GOV-01

Every architecture component must map to requirements.

RTM-ARCH-GOV-02

Every architecture component must be testable.

RTM-ARCH-GOV-03

Every architecture component must be governed.

RTM-ARCH-GOV-04

Every architecture component must support audits.

RTM-ARCH-GOV-05

Coverage must be measurable.

RTM-ARCH-GOV-06

Release validation is mandatory.

6.28 Architectural Constraints

AC-RTM-ARCH-01

Architecture without requirements is prohibited.

AC-RTM-ARCH-02

Architecture without testing is prohibited.

AC-RTM-ARCH-03

Architecture without governance validation is prohibited.

AC-RTM-ARCH-04

Traceability evidence cannot be deleted.

AC-RTM-ARCH-05

Coverage gaps require remediation.

AC-RTM-ARCH-06

Release approval requires complete architecture traceability.

6.29 Section Summary

This section defines the Architecture Traceability Matrix and establishes mappings between system requirements and architectural components throughout TODP.

The Architecture Traceability Matrix ensures that every architectural decision, service, infrastructure component, governance control, and operational capability can be traced back to approved requirements and forward into implementation, testing, governance validation, auditing, and release approval.

RTM V1 Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Framework

☐ Section 3 — Requirement Classification Model

☐ Section 4 — BRD Traceability Matrix

☐ Section 5 — SRS Traceability Matrix

☐ Section 6 — Architecture Traceability Matrix

☐ Section 7 — Component Traceability Matrix

RTM V1 — Section 7

Component Traceability Matrix

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

Version:

V1

Status:

APPROVED

7.1 Purpose

This section defines the Component Traceability Matrix for TODP.

The Component Traceability Matrix establishes traceability between architectural components, technical components, implementation artifacts, verification artifacts, governance controls, and release artifacts.

The objective is to ensure:

- Component accountability
- Requirement realization
- Verification coverage
- Governance alignment
- Audit readiness
- Release readiness

7.2 Component Traceability Objectives

RTM-COMP-01

Verify component ownership.

RTM-COMP-02

Verify component coverage.

RTM-COMP-03

Verify implementation alignment.

RTM-COMP-04

Verify governance alignment.

RTM-COMP-05

Verify auditability.

RTM-COMP-06

Verify release readiness.

7.3 Component Traceability Model

Requirement



Architecture Component



Technical Component



Source Code Component



API Component



Database Component



Test Component



Governance Control



Release Artifact

7.4 TODP Component Inventory

Core Components:

Authentication Component

Authorization Component

Opportunity Component

Authority Component

Evidence Component

Retrieval Component

Ranking Component

Recommendation Component

Governance Component

Audit Component

Replay Component

Observability Component

7.5 Authentication Component Traceability

Component ID:

COMP-AUTH-001

Authentication Component

Maps To:

SRS-NFR-SEC-001

Authentication Requirements



SAD-AUTH-001

Authentication Service



TDD-AUTH

Authentication Design



API-AUTHN

Authentication APIs



TEST-AUTH

Authentication Tests



GOV-SEC

Security Governance



REL-AUTH

Release Validation

7.6 Authorization Component Traceability

Component ID:

COMP-RBAC-001

Authorization Component

Maps To:

SRS-NFR-SEC-002

Authorization Requirements



SAD-RBAC-001

Authorization Service



TDD-RBAC

RBAC Design



API-RBAC

Authorization APIs



TEST-RBAC

Authorization Tests



GOV-RBAC

Governance Validation



REL-RBAC

Release Validation

7.7 Opportunity Component Traceability

Component ID:

COMP-OPP-001

Opportunity Component

Maps To:

BRD-OPP-001

Opportunity Management



SRS-FR-OPP-001

Opportunity Requirements



SAD-COMP-OPP

Opportunity Service



TDD-OPP

Opportunity Design



TEST-OPP

Opportunity Tests



GOV-OPP

Governance Controls



REL-OPP

Release Validation

7.8 Authority Component Traceability

Component ID:

COMP-AUTHORITY-001

Authority Component

Maps To:

BRD-AUTH-001

Authority Management



SRS-FR-AUTH-001

Authority Requirements



SAD-COMP-AUTH

Authority Service



TDD-AUTH

Authority Design



TEST-AUTH

Authority Tests



GOV-AUTH

Governance Controls



REL-AUTH

Release Validation

7.9 Evidence Component Traceability

Component ID:

COMP-EVIDENCE-001

Evidence Component

Maps To:

BRD-EVD-001

Evidence Management



SRS-FR-EVD-001

Evidence Requirements



SAD-COMP-EVD

Evidence Service



TDD-EVD

Evidence Design



TEST-EVD

Evidence Tests



GOV-EVD

Governance Controls



REL-EVD

Release Validation

7.10 Retrieval Component Traceability

Component ID:

COMP-RET-001

Retrieval Component

Maps To:

BRD-RET-001

Retrieval Requirements



SRS-FR-RET-001

Hybrid Retrieval



SAD-COMP-RET

Retrieval Engine



TDD-RET

Retrieval Design



TEST-RET

Retrieval Tests



GOV-RET

Governance Controls



REL-RET

Release Validation

7.11 Ranking Component Traceability

Component ID:

COMP-RANK-001

Ranking Component

Maps To:

BRD-RANK-001

Ranking Requirements



SRS-FR-RANK-001

Ranking Execution



SAD-COMP-RANK

Ranking Engine



TDD-RANK

Ranking Design



TEST-RANK

Ranking Tests



GOV-RANK

Governance Controls



REL-RANK

Release Validation

7.12 Recommendation Component Traceability

Component ID:

COMP-REC-001

Recommendation Component

Maps To:

BRD-REC-001

Recommendation Requirements



SRS-FR-REC-001

Recommendation Generation



SAD-COMP-REC

Recommendation Engine



TDD-REC

Recommendation Design



TEST-REC

Recommendation Tests



GOV-REC

Governance Controls



REL-REC

Release Validation

7.13 Governance Component Traceability

Component ID:

COMP-GOV-001

Governance Component

Maps To:

BRD-GOV-001

Governance Requirements



SRS-FR-GOV-001

Governance Workflows



SAD-COMP-GOV

Governance Service



TDD-GOV

Governance Design



TEST-GOV

Governance Tests



GOV-GOV

Governance Controls



REL-GOV

Release Validation

7.14 Audit Component Traceability

Component ID:

COMP-AUD-001

Audit Component

Maps To:

BRD-AUD-001

Audit Requirements



SRS-FR-AUD-001

Audit Logging



SAD-COMP-AUD

Audit Service



TDD-AUD

Audit Design



TEST-AUD

Audit Tests



GOV-AUD

Governance Controls



REL-AUD

Release Validation

7.15 Replay Component Traceability

Component ID:

COMP-REP-001

Replay Component

Maps To:

BRD-REP-001

Replay Requirements



SRS-FR-REP-001

Replay Execution



SAD-COMP-REP

Replay Engine



TDD-REP

Replay Design



TEST-REP

Replay Tests



GOV-REP

Governance Controls



REL-REP

Release Validation

7.16 Search Component Traceability

Component ID:

COMP-SEARCH-001

Hybrid Search Component

Maps To:

SRS-FR-RET-001

Search Requirements



SAD-INFRA-SEARCH

Search Architecture



TDD-SEARCH

Search Design



TEST-SEARCH

Search Tests



REL-SEARCH

Release Validation

7.17 Database Component Traceability

Component ID:

COMP-DB-001

Database Component

Maps To:

Persistence Requirements

↓

SAD-INFRA-DB

Database Architecture

↓

TDD-DATABASE

Database Design



TEST-DB

Database Tests



GOV-DATA

Data Governance



REL-DB

Release Validation

7.18 Cache Component Traceability

Component ID:

COMP-CACHE-001

Cache Component

Maps To:

Performance Requirements



SAD-INFRA-CACHE

Cache Architecture



TDD-CACHE

Cache Design



TEST-CACHE

Cache Tests



REL-CACHE

Release Validation

7.19 Observability Component Traceability

Component ID:

COMP-OBS-001

Observability Component

Maps To:

Operational Requirements



SAD-OBS-001

Observability Architecture



TDD-OBS

Observability Design



TEST-OBS

Observability Tests



GOV-OBS

Operational Governance



REL-OBS

Release Validation

7.20 Infrastructure Component Traceability

Component ID:

COMP-INFRA-001

Infrastructure Component

Maps To:

Operational Requirements



SAD-INFRA

Infrastructure Architecture



TDD-INFRA

Infrastructure Design



TEST-INFRA

Infrastructure Tests



GOV-INFRA

Infrastructure Governance



REL-INFRA

Release Validation

7.21 Component Ownership Requirements

Each component shall have:

Business Owner

Technical Owner

QA Owner

Governance Owner

Release Owner

7.22 Component Verification Requirements

Each component shall map to:

Unit Tests

Integration Tests

API Tests

Security Tests

Performance Tests

UAT Tests

7.23 Component Governance Requirements

Each component shall support:

Governance Reviews

Compliance Reviews

Audit Reviews

Security Reviews

Release Reviews

7.24 Component Coverage Requirements

Every component shall map to:

Requirement

Architecture

Design

Implementation

Verification

Governance

Release

7.25 Component Gap Analysis

Component gaps exist when:

No Requirement Mapping

No Verification Mapping

No Governance Mapping

No Release Mapping

Missing Ownership

All gaps require remediation.

7.26 Component Impact Analysis

Changes to components shall trigger review of:

Requirements

Architecture

Design

Database

APIs

Tests

Governance Controls

Release Plans

7.27 Coverage Metrics

Required Metrics:

Component Coverage %

Verification Coverage %

Governance Coverage %

Release Coverage %

Audit Coverage %

Target:

100%

7.28 Governance Rules

RTM-COMP-GOV-01

Every component must have ownership.

RTM-COMP-GOV-02

Every component must be traceable.

RTM-COMP-GOV-03

Every component must be testable.

RTM-COMP-GOV-04

Every component must support governance validation.

RTM-COMP-GOV-05

Coverage must be measurable.

RTM-COMP-GOV-06

Release validation is mandatory.

7.29 Architectural Constraints

AC-RTM-COMP-01

Unmapped components are prohibited.

AC-RTM-COMP-02

Components without testing are prohibited.

AC-RTM-COMP-03

Components without governance validation are prohibited.

AC-RTM-COMP-04

Ownership is mandatory.

AC-RTM-COMP-05

Coverage gaps require remediation.

AC-RTM-COMP-06

Release approval requires complete component traceability.

7.30 Section Summary

This section defines the Component Traceability Matrix and establishes mappings between TODP components, requirements, architectural services, technical designs, verification artifacts, governance controls, and release approvals.

The Component Traceability Matrix ensures that every major platform component remains traceable, verifiable, governable, auditable, and release-ready throughout the system lifecycle.

RTM V1 Progress

▢ Section 1 — Document Overview & Traceability Scope

▢ Section 2 — Traceability Framework

▢ Section 3 — Requirement Classification Model

▢ Section 4 — BRD Traceability Matrix

▢ Section 5 — SRS Traceability Matrix

▢ Section 6 — Architecture Traceability Matrix

▢ Section 7 — Component Traceability Matrix

▢ Section 8 — Database Traceability Matrix

RTM V1 — Section 8

Database Traceability Matrix

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

8

Version:

V1

Status:

APPROVED

8.1 Purpose

This section defines the Database Traceability Matrix for TODP.

The Database Traceability Matrix establishes traceability between business requirements, system requirements, database entities, persistence models, APIs, verification artifacts, governance controls, and release artifacts.

The objective is to ensure:

- Data accountability
- Requirement realization
- Data governance compliance
- Auditability
- Verification coverage
- Release readiness

8.2 Database Traceability Objectives

RTM-DB-01

Verify database coverage.

RTM-DB-02

Verify requirement realization.

RTM-DB-03

Verify data ownership.

RTM-DB-04

Verify governance compliance.

RTM-DB-05

Verify auditability.

RTM-DB-06

Verify release readiness.

8.3 Database Traceability Model

Business Requirement



System Requirement



Architecture Component



Database Entity



Repository Layer



API Layer



Test Artifact



Governance Control



Release Artifact

8.4 Database Domains

The following domains shall be traced:

Identity Domain

Opportunity Domain

Authority Domain

Evidence Domain

Retrieval Domain

Ranking Domain

Recommendation Domain

Governance Domain

Audit Domain

Replay Domain

Operational Domain

8.5 User Entity Traceability

Database Entity:

DB-USER

Users Table

Maps To:

BRD-SEC-001

Identity Management



SRS-FR-USER-001

User Management



SAD-AUTH-001

Authentication Service



TDD-USER

User Design



API-USER

User APIs



TEST-USER

User Tests



GOV-SEC

Security Governance



REL-USER

Release Validation

8.6 Role Entity Traceability

Database Entity:

DB-ROLE

Roles Table

Maps To:

BRD-SEC-002

Access Control



SRS-FR-RBAC-001

Role Management



SAD-RBAC-001

Authorization Service



TDD-RBAC

RBAC Design



API-RBAC

RBAC APIs



TEST-RBAC

RBAC Tests



GOV-RBAC

Governance Controls



REL-RBAC

Release Validation

8.7 Opportunity Entity Traceability

Database Entity:

DB-OPPORTUNITY

Opportunity Table

Maps To:

BRD-OPP-001

Opportunity Management

↓

SRS-FR-OPP-001

Opportunity CRUD

↓

SAD-COMP-OPP

Opportunity Service

↓

TDD-OPP

Opportunity Design



API-OPP

Opportunity APIs



TEST-OPP

Opportunity Tests



GOV-OPP

Governance Controls



REL-OPP

Release Validation

8.8 Authority Entity Traceability

Database Entity:

DB-AUTHORITY

Authority Table

Maps To:

BRD-AUTH-001

Authority Management

↓

SRS-FR-AUTH-001

Authority CRUD

↓

SAD-COMP-AUTH

Authority Service



TDD-AUTH

Authority Design



API-AUTH

Authority APIs



TEST-AUTH

Authority Tests



GOV-AUTH

Governance Controls



REL-AUTH

Release Validation

8.9 Evidence Entity Traceability

Database Entity:

DB-EVIDENCE

Evidence Table

Maps To:

BRD-EVD-001

Evidence Management



SRS-FR-EVD-001

Evidence CRUD



SAD-COMP-EVD

Evidence Service



TDD-EVD

Evidence Design



API-EVD

Evidence APIs



TEST-EVD

Evidence Tests



GOV-EVD

Governance Controls



REL-EVD

Release Validation

8.10 Evidence Link Entity Traceability

Database Entity:

DB-EVIDENCE_LINK

Evidence Relationship Table

Maps To:

BRD-EVD-002

Evidence Lineage



SRS-FR-EVD-002

Evidence Relationships



SAD-COMP-EVD

Evidence Service



TDD-EVD-LINK

Evidence Link Design



API-EVD

Evidence APIs



TEST-EVD

Evidence Tests



GOV-EVD

Governance Controls



REL-EVD

Release Validation

8.11 Retrieval History Entity Traceability

Database Entity:

DB-RETRIEVAL_HISTORY

Retrieval History Table

Maps To:

BRD-RET-001

Search Requirements

↓

SRS-FR-RET-001

Retrieval Execution

↓

SAD-COMP-RET

Retrieval Engine

↓

TDD-RET

Retrieval Design



API-RET

Retrieval APIs



TEST-RET

Retrieval Tests



GOV-RET

Governance Controls



REL-RET

Release Validation

8.12 Embedding Entity Traceability

Database Entity:

DB-EMBEDDING

Embedding Store

Maps To:

BRD-RET-002

Semantic Search

↓

SRS-FR-RET-002

Embedding Requirements

↓

SAD-INFRA-SEARCH

Search Infrastructure



TDD-SEARCH

Search Design



API-RET

Retrieval APIs



TEST-SEARCH

Search Tests



GOV-RET

Governance Controls



REL-SEARCH

Release Validation

8.13 Ranking Entity Traceability

Database Entity:

DB-RANKING

Ranking Table

Maps To:

BRD-RANK-001

Ranking Requirements

↓

SRS-FR-RANK-001

Ranking Execution



SAD-COMP-RANK

Ranking Engine



TDD-RANK

Ranking Design



API-RANK

Ranking APIs



TEST-RANK

Ranking Tests



GOV-RANK

Governance Controls

↓

REL-RANK

Release Validation

8.14 Recommendation Entity Traceability

Database Entity:

DB-RECOMMENDATION

Recommendation Table

Maps To:

BRD-REC-001

Recommendation Requirements



SRS-FR-REC-001

Recommendation Execution



SAD-COMP-REC

Recommendation Engine



TDD-REC

Recommendation Design



API-REC

Recommendation APIs



TEST-REC

Recommendation Tests

↓

GOV-REC

Governance Controls

↓

REL-REC

Release Validation

8.15 Governance Entity Traceability

Database Entity:

DB-GOVERNANCE

Governance Table

Maps To:

BRD-GOV-001

Governance Requirements



SRS-FR-GOV-001

Policy Enforcement



SAD-COMP-GOV

Governance Service



TDD-GOV

Governance Design



API-GOV

Governance APIs



TEST-GOV

Governance Tests



GOV-GOV

Governance Controls



REL-GOV

Release Validation

8.16 Audit Entity Traceability

Database Entity:

DB-AUDIT

Audit Table

Maps To:

BRD-AUD-001

Audit Requirements

↓

SRS-FR-AUD-001

Audit Logging

↓

SAD-COMP-AUD

Audit Service

↓

TDD-AUD

Audit Design



API-AUD

Audit APIs



TEST-AUD

Audit Tests



GOV-AUD

Governance Controls



REL-AUD

Release Validation

8.17 Snapshot Entity Traceability

Database Entity:

DB-SNAPSHOT

Replay Snapshot Table

Maps To:

BRD-REP-001

Replay Requirements

↓

SRS-FR-REP-001

Replay Execution

↓

SAD-COMP-REP

Replay Engine



TDD-REP

Replay Design



API-REP

Replay APIs



TEST-REP

Replay Tests



GOV-REP

Governance Controls



REL-REP

Release Validation

8.18 Configuration Entity Traceability

Database Entity:

DB-CONFIG

Configuration Table

Maps To:

Operational Requirements



SRS-OPS-001

Configuration Management



SAD-INFRA

Infrastructure Architecture



TDD-CONFIG

Configuration Design



API-CONFIG

Configuration APIs



TEST-CONFIG

Configuration Tests



GOV-INFRA

Infrastructure Governance



REL-INFRA

Release Validation

8.19 Database Relationship Traceability

All database relationships shall trace to:

Business Rules

System Requirements

Architecture Components

Technical Designs

Verification Artifacts

Governance Controls

8.20 Referential Integrity Traceability

Referential integrity controls shall map to:

Entity Relationships

↓

Business Rules

↓

Data Validation Rules

↓

Test Cases

↓

Governance Validation

8.21 Repository Traceability Requirements

Every database entity shall map to:

Repository Layer

Service Layer

API Layer

Verification Layer

Governance Layer

Release Layer

8.22 Data Governance Traceability

Each entity shall support:

Ownership

Classification

Retention Rules

Access Controls

Audit Controls

Compliance Controls

8.23 Database Coverage Requirements

Every database entity shall map to:

Requirement

Architecture

Design

API

Testing

Governance

Release

8.24 Database Gap Analysis

Database gaps exist when:

Entity Without Requirement

Requirement Without Entity

Missing Tests

Missing Governance Controls

Missing Release Validation

All gaps require remediation.

8.25 Database Impact Analysis

Changes to database entities shall trigger review of:

Requirements

Architecture

Design

Repositories

APIs

Tests

Governance Controls

Release Plans

8.26 Coverage Metrics

Required Metrics:

Database Coverage %

Verification Coverage %

Governance Coverage %

Release Coverage %

Audit Coverage %

Target:

100%

8.27 Governance Rules

RTM-DB-GOV-01

Every database entity must be traceable.

RTM-DB-GOV-02

Every database entity must be tested.

RTM-DB-GOV-03

Every database entity must support governance validation.

RTM-DB-GOV-04

Every database entity must support audits.

RTM-DB-GOV-05

Coverage must be measurable.

RTM-DB-GOV-06

Release validation is mandatory.

8.28 Architectural Constraints

AC-RTM-DB-01

Database entities without requirements are prohibited.

AC-RTM-DB-02

Database entities without testing are prohibited.

AC-RTM-DB-03

Database entities without governance validation are prohibited.

AC-RTM-DB-04

Coverage gaps require remediation.

AC-RTM-DB-05

Traceability evidence cannot be deleted.

AC-RTM-DB-06

Release approval requires complete database traceability.

8.29 Section Summary

This section defines the Database Traceability Matrix and establishes mappings between database entities, requirements, architecture components, technical designs, APIs, verification artifacts, governance controls, and release approvals.

The Database Traceability Matrix ensures that every persistence artifact within TODP remains traceable, governed, auditable, testable, and release-ready throughout its lifecycle.

RTM V1 Progress

- ❏ Section 2 — Traceability Framework
- ❏ Section 3 — Requirement Classification Model
- ❏ Section 4 — BRD Traceability Matrix
- ❏ Section 5 — SRS Traceability Matrix
- ❏ Section 6 — Architecture Traceability Matrix
- ❏ Section 7 — Component Traceability Matrix
- ❏ Section 8 — Database Traceability Matrix
- ❏ Section 9 — API Traceability Matrix

RTM V1 — Section 9

API Traceability Matrix

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

9

Version:

V1

Status:

APPROVED

9.1 Purpose

This section defines the API Traceability Matrix for TODP.

The API Traceability Matrix establishes traceability between business requirements, system requirements, architecture components, API contracts, database entities, verification artifacts, governance controls, and release artifacts.

The objective is to ensure:

- API accountability
- Requirement realization
- Contract consistency
- Verification coverage
- Governance compliance
- Release readiness

9.2 API Traceability Objectives

RTM-API-01

Verify API coverage.

RTM-API-02

Verify requirement realization.

RTM-API-03

Verify contract traceability.

RTM-API-04

Verify governance compliance.

RTM-API-05

Verify auditability.

RTM-API-06

Verify release readiness.

9.3 API Traceability Model

Business Requirement



System Requirement



Architecture Component



API Contract



Service Component



Database Entity



Test Artifact



Governance Control



Release Artifact

9.4 API Domains

The following API domains shall be traced:

Authentication APIs

Authorization APIs

Opportunity APIs

Authority APIs

Evidence APIs

Retrieval APIs

Ranking APIs

Recommendation APIs

Governance APIs

Audit APIs

Replay APIs

Administrative APIs

9.5 Authentication API Traceability

API ID:

API-AUTHN-001

Login API

Maps To:

BRD-SEC-001

Identity Management

↓

SRS-NFR-SEC-001

Authentication Requirements



SAD-AUTH-001

Authentication Service



COMP-AUTH-001

Authentication Component



DB-USER

User Entity



TEST-AUTH-001

Authentication Tests



GOV-SEC-001

Security Governance



REL-AUTH-001

Release Validation

9.6 Authorization API Traceability

API ID:

API-RBAC-001

Role Management API

Maps To:

BRD-SEC-002

Access Control



SRS-FR-RBAC-001

Authorization Requirements



SAD-RBAC-001

Authorization Service



COMP-RBAC-001

Authorization Component



DB-ROLE

Role Entity



TEST-RBAC-001

RBAC Tests



GOV-RBAC-001

RBAC Governance



REL-RBAC-001

Release Validation

9.7 Opportunity API Traceability

API ID:

API-OPP-001

Opportunity CRUD APIs

Maps To:

BRD-OPP-001

Opportunity Management

↓

SRS-FR-OPP-001

Opportunity CRUD

↓

SAD-COMP-OPP

Opportunity Service

↓

COMP-OPP-001

Opportunity Component



DB-OPPORTUNITY

Opportunity Entity



TEST-OPP-001

Opportunity Tests



GOV-OPP-001

Governance Controls



REL-OPP-001

Release Validation

9.8 Authority API Traceability

API ID:

API-AUTH-001

Authority CRUD APIs

Maps To:

BRD-AUTH-001

Authority Management

↓

SRS-FR-AUTH-001

Authority CRUD

↓

SAD-COMP-AUTH

Authority Service



COMP-AUTHORITY-001

Authority Component



DB-AUTHORITY

Authority Entity



TEST-AUTH-001

Authority Tests



GOV-AUTH-001

Governance Controls



REL-AUTH-001

Release Validation

9.9 Evidence API Traceability

API ID:

API-EVD-001

Evidence CRUD APIs

Maps To:

BRD-EVD-001

Evidence Management

↓

SRS-FR-EVD-001

Evidence CRUD



SAD-COMP-EVD

Evidence Service



COMP-EVIDENCE-001

Evidence Component



DB-EVIDENCE

Evidence Entity



TEST-EVD-001

Evidence Tests



GOV-EVD-001

Governance Controls



REL-EVD-001

Release Validation

9.10 Retrieval API Traceability

API ID:

API-RET-001

Hybrid Retrieval APIs

Maps To:

BRD-RET-001

Search Requirements



SRS-FR-RET-001

Hybrid Retrieval



SAD-COMP-RET

Retrieval Engine



COMP-RET-001

Retrieval Component



DB-RETRIEVAL_HISTORY

Retrieval Entity



TEST-RET-001

Retrieval Tests



GOV-RET-001

Governance Controls



REL-RET-001

Release Validation

9.11 Ranking API Traceability

API ID:

API-RANK-001

Ranking APIs

Maps To:

BRD-RANK-001

Ranking Requirements



SRS-FR-RANK-001

Ranking Execution



SAD-COMP-RANK

Ranking Engine



COMP-RANK-001

Ranking Component



DB-RANKING

Ranking Entity



TEST-RANK-001

Ranking Tests



GOV-RANK-001

Governance Controls



REL-RANK-001

Release Validation

9.12 Recommendation API Traceability

API ID:

API-REC-001

Recommendation APIs

Maps To:

BRD-REC-001

Recommendation Requirements

↓

SRS-FR-REC-001

Recommendation Generation

↓

SAD-COMP-REC

Recommendation Engine

↓

COMP-REC-001

Recommendation Component



DB-RECOMMENDATION

Recommendation Entity



TEST-REC-001

Recommendation Tests



GOV-REC-001

Governance Controls



REL-REC-001

Release Validation

9.13 Governance API Traceability

API ID:

API-GOV-001

Governance APIs

Maps To:

BRD-GOV-001

Governance Requirements

↓

SRS-FR-GOV-001

Policy Enforcement

↓

SAD-COMP-GOV

Governance Service



COMP-GOV-001

Governance Component



DB-GOVERNANCE

Governance Entity



TEST-GOV-001

Governance Tests



GOV-GOV-001

Governance Controls



REL-GOV-001

Release Validation

9.14 Audit API Traceability

API ID:

API-AUD-001

Audit APIs

Maps To:

BRD-AUD-001

Audit Requirements



SRS-FR-AUD-001

Audit Logging



SAD-COMP-AUD

Audit Service



COMP-AUD-001

Audit Component



DB-AUDIT

Audit Entity



TEST-AUD-001

Audit Tests



GOV-AUD-001

Governance Controls



REL-AUD-001

Release Validation

9.15 Replay API Traceability

API ID:

API-REP-001

Replay APIs

Maps To:

BRD-REP-001

Replay Requirements



SRS-FR-REP-001

Replay Execution



SAD-COMP-REP

Replay Engine



COMP-REP-001

Replay Component



DB-SNAPSHOT

Snapshot Entity



TEST-REP-001

Replay Tests



GOV-REP-001

Governance Controls



REL-REP-001

Release Validation

9.16 Administrative API Traceability

API ID:

API-ADMIN-001

Administration APIs

Maps To:

Operational Requirements



SRS-OPS-001

Administration Requirements



SAD-INFRA

Infrastructure Services



COMP-INFRA-001

Infrastructure Component



DB-CONFIG

Configuration Entity



TEST-ADMIN-001

Administration Tests



GOV-INFRA-001

Infrastructure Governance



REL-INFRA-001

Release Validation

9.17 API Contract Traceability

Every API contract shall map to:

Requirement

Architecture Component

Service Component

Database Entity

Verification Artifact

Governance Control

Release Artifact

9.18 Request Model Traceability

Request models shall trace to:

Business Rules

Validation Rules

Security Rules

Governance Rules

Testing Artifacts

9.19 Response Model Traceability

Response models shall trace to:

Business Requirements

Functional Requirements

Audit Requirements

Governance Requirements

Verification Artifacts

9.20 API Security Traceability

Security controls shall trace to:

Authentication Requirements

Authorization Requirements

Audit Requirements

Governance Controls

Security Tests

9.21 API Versioning Traceability

Every API version shall maintain:

Version Identifier

Requirement Mapping

Change History

Approval History

Release History

9.22 API Coverage Requirements

Every API shall map to:

Requirement

Architecture

Component

Database Entity

Verification

Governance

Release

9.23 API Gap Analysis

API gaps exist when:

Requirement Missing API

API Missing Requirement

Missing Tests

Missing Governance Validation

Missing Release Validation

All gaps require remediation.

9.24 API Impact Analysis

Changes to APIs shall trigger review of:

Requirements

Architecture

Components

Database Entities

Test Cases

Governance Controls

Release Plans

9.25 Coverage Metrics

Required Metrics:

API Coverage %

Verification Coverage %

Governance Coverage %

Release Coverage %

Audit Coverage %

Target:

100%

9.26 Governance Rules

RTM-API-GOV-01

Every API must be traceable.

RTM-API-GOV-02

Every API must be tested.

RTM-API-GOV-03

Every API must support governance validation.

RTM-API-GOV-04

Every API must support audits.

RTM-API-GOV-05

Coverage must be measurable.

RTM-API-GOV-06

Release validation is mandatory.

9.27 Architectural Constraints

AC-RTM-API-01

APIs without requirements are prohibited.

AC-RTM-API-02

APIs without testing are prohibited.

AC-RTM-API-03

APIs without governance validation are prohibited.

AC-RTM-API-04

Coverage gaps require remediation.

AC-RTM-API-05

Traceability evidence cannot be deleted.

AC-RTM-API-06

Release approval requires complete API traceability.

9.28 Section Summary

This section defines the API Traceability Matrix and establishes mappings between API contracts, business requirements, system requirements, architecture components, service components, database entities, verification artifacts, governance controls, and release approvals.

The API Traceability Matrix ensures that every API exposed by TODP remains traceable, testable, governed, auditable, version-controlled, and release-ready throughout its lifecycle.

RTM V1 Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Framework

☐ Section 3 — Requirement Classification Model

☐ Section 4 — BRD Traceability Matrix

☐ Section 5 — SRS Traceability Matrix

☐ Section 6 — Architecture Traceability Matrix

☐ Section 7 — Component Traceability Matrix

☐ Section 8 — Database Traceability Matrix

☐ Section 9 — API Traceability Matrix

☐ Section 10 — Opportunity Module Traceability

RTM V1 — Section 10

Opportunity Module Traceability

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

10

Version:

V1

Status:

APPROVED

10.1 Purpose

This section defines the Opportunity Module Traceability Matrix for TODP.

The Opportunity Module serves as the primary business object of the platform and represents opportunities that are evaluated, ranked, governed, recommended, audited, and replayed.

The objective is to ensure:

- Complete opportunity lifecycle traceability
- Business requirement realization
- Data accountability
- Verification coverage
- Governance compliance
- Release readiness

10.2 Opportunity Traceability Objectives

RTM-OPP-01

Verify opportunity requirement coverage.

RTM-OPP-02

Verify implementation traceability.

RTM-OPP-03

Verify governance compliance.

RTM-OPP-04

Verify auditability.

RTM-OPP-05

Verify replayability.

RTM-OPP-06

Verify release readiness.

10.3 Opportunity Traceability Model

Business Requirement



System Requirement



Architecture Component



Technical Design



API Contract



Database Entity



Verification Artifact



Governance Control



Audit Artifact



Release Artifact

10.4 Opportunity Business Requirements

Primary Business Requirements:

BRD-OPP-001

Manage Opportunities

BRD-OPP-002

Track Opportunity Lifecycle

BRD-OPP-003

Support Opportunity Discovery

BRD-OPP-004

Support Opportunity Evaluation

BRD-OPP-005

Support Opportunity Governance

10.5 Opportunity Functional Requirements

Mapped Requirements:

SRS-FR-OPP-001

Create Opportunity

SRS-FR-OPP-002

Update Opportunity

SRS-FR-OPP-003

Delete Opportunity

SRS-FR-OPP-004

Search Opportunities

SRS-FR-OPP-005

View Opportunity History

10.6 Opportunity Architecture Traceability

Architecture Component:

SAD-COMP-OPP

Opportunity Service

Responsibilities:

Opportunity Management

Opportunity Validation

Opportunity Lifecycle

Opportunity Governance

Opportunity Auditability

10.7 Opportunity Design Traceability

Technical Design:

TDD-OPP

Opportunity Design Module

Includes:

Domain Models

Validation Rules

Business Rules

Workflow Logic

Persistence Logic

10.8 Opportunity Database Traceability

Database Entity:

DB-OPPORTUNITY

Opportunity Table

Stores:

Opportunity Metadata

Status

Ownership

Lifecycle State

Governance State

Audit References

10.9 Opportunity API Traceability

API Group:

API-OPP

Opportunity APIs

Includes:

POST /opportunities

GET /opportunities

GET /opportunities/{id}

PUT /opportunities/{id}

DELETE /opportunities/{id}

10.10 Opportunity Component Traceability

Component:

COMP-OPP-001

Opportunity Component

Maps To:

Opportunity Service

Opportunity APIs

Opportunity Repository

Opportunity Validators

Opportunity Governance Controls

10.11 Opportunity Repository Traceability

Repository Layer:

REPO-OPP

Opportunity Repository

Responsibilities:

Persistence

Retrieval

Versioning

Lifecycle Tracking

Audit Linking

10.12 Opportunity Validation Traceability

Validation Rules:

VAL-OPP-001

Required Fields

VAL-OPP-002

Status Validation

VAL-OPP-003

Ownership Validation

VAL-OPP-004

Governance Validation

10.13 Opportunity Lifecycle Traceability

Lifecycle States:

Draft



Review



Approved



Active



Completed



Archived

Each state change must be traceable.

10.14 Opportunity Governance Traceability

Governance Controls:

GOV-OPP-001

Opportunity Approval

GOV-OPP-002

Policy Enforcement

GOV-OPP-003

Compliance Validation

GOV-OPP-004

Lifecycle Governance

10.15 Opportunity Audit Traceability

Audit Controls:

AUD-OPP-001

Creation Audit

AUD-OPP-002

Modification Audit

AUD-OPP-003

Deletion Audit

AUD-OPP-004

Governance Audit

AUD-OPP-005

Replay Audit

10.16 Opportunity Replay Traceability

Replay Controls:

REP-OPP-001

Historical Replay

REP-OPP-002

State Reconstruction

REP-OPP-003

Decision Replay

REP-OPP-004

Governance Replay

10.17 Opportunity Testing Traceability

Verification Artifacts:

TEST-OPP-UNIT

Unit Tests

TEST-OPP-INT

Integration Tests

TEST-OPP-API

API Tests

TEST-OPP-SEC

Security Tests

TEST-OPP-UAT

UAT Tests

10.18 Opportunity Security Traceability

Security Controls:

SEC-OPP-001

Authentication

SEC-OPP-002

Authorization

SEC-OPP-003

Ownership Controls

SEC-OPP-004

Audit Logging

10.19 Opportunity Data Governance Traceability

Data Governance Controls:

DATA-OPP-001

Ownership Rules

DATA-OPP-002

Retention Rules

DATA-OPP-003

Classification Rules

DATA-OPP-004

Access Controls

10.20 Opportunity Reporting Traceability

Reports Supported:

Opportunity Summary Report

Opportunity Lifecycle Report

Governance Report

Audit Report

Compliance Report

10.21 Opportunity Release Traceability

Release Controls:

REL-OPP-001

Functional Validation

REL-OPP-002

Security Validation

REL-OPP-003

Governance Validation

REL-OPP-004

Release Approval

10.22 Opportunity Coverage Requirements

Every opportunity capability shall map to:

Requirement

Architecture

Design

Database

API

Testing

Governance

Audit

Release

10.23 Opportunity Gap Analysis

Opportunity gaps exist when:

Missing Requirement Mapping

Missing API Mapping

Missing Database Mapping

Missing Test Coverage

Missing Governance Validation

Missing Release Validation

All gaps require remediation.

10.24 Opportunity Impact Analysis

Changes to opportunity requirements shall trigger review of:

Architecture

Design

Database

APIs

Tests

Governance Controls

Audit Controls

Release Plans

10.25 Coverage Metrics

Required Metrics:

Opportunity Requirement Coverage %

Opportunity Test Coverage %

Opportunity Governance Coverage %

Opportunity Audit Coverage %

Opportunity Release Coverage %

Target:

100%

10.26 Governance Rules

RTM-OPP-GOV-01

Every opportunity capability must be traceable.

RTM-OPP-GOV-02

Every opportunity capability must be tested.

RTM-OPP-GOV-03

Every opportunity capability must support governance validation.

RTM-OPP-GOV-04

Every opportunity capability must support audits.

RTM-OPP-GOV-05

Coverage must be measurable.

RTM-OPP-GOV-06

Release validation is mandatory.

10.27 Architectural Constraints

AC-RTM-OPP-01

Opportunity features without requirements are prohibited.

AC-RTM-OPP-02

Opportunity features without testing are prohibited.

AC-RTM-OPP-03

Opportunity features without governance validation are prohibited.

AC-RTM-OPP-04

Coverage gaps require remediation.

AC-RTM-OPP-05

Traceability evidence cannot be deleted.

AC-RTM-OPP-06

Release approval requires complete opportunity traceability.

10.28 Section Summary

This section defines the Opportunity Module Traceability Matrix and establishes mappings between business requirements, system requirements, architecture components, technical designs, APIs, database entities, verification artifacts, governance controls, audit controls, replay controls, and release approvals.

The Opportunity Module Traceability Matrix ensures that the core business object of TODP remains fully traceable, governable, auditable, replayable, testable, and release-ready throughout its lifecycle.

RTM V1 Progress

🔗 Section 1 — Document Overview & Traceability Scope

🔗 Section 2 — Traceability Framework

🔗 Section 3 — Requirement Classification Model

🔗 Section 4 — BRD Traceability Matrix

🔗 Section 5 — SRS Traceability Matrix

🔗 Section 6 — Architecture Traceability Matrix

🔗 Section 7 — Component Traceability Matrix

🔗 Section 8 — Database Traceability Matrix

🔗 Section 9 — API Traceability Matrix

🔗 Section 10 — Opportunity Module Traceability

🔗 Section 11 — Authority Module Traceability

RTM V1 — Section 11

Authority Module Traceability

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

11

Version:

V1

Status:

APPROVED

11.1 Purpose

This section defines the Authority Module Traceability Matrix for TODP.

The Authority Module manages authority sources, authority credibility, authority validation, authority lineage, and authority influence within recommendation and governance workflows.

The objective is to ensure:

- Authority accountability
- Authority transparency
- Authority governance
- Authority auditability
- Authority verification
- Release readiness

11.2 Authority Traceability Objectives

RTM-AUTH-01

Verify authority requirement coverage.

RTM-AUTH-02

Verify authority implementation traceability.

RTM-AUTH-03

Verify authority governance compliance.

RTM-AUTH-04

Verify authority auditability.

RTM-AUTH-05

Verify authority lineage traceability.

RTM-AUTH-06

Verify release readiness.

11.3 Authority Traceability Model

Business Requirement



System Requirement



Architecture Component



Technical Design



API Contract



Database Entity



Verification Artifact



Governance Control



Audit Artifact



Release Artifact

11.4 Authority Business Requirements

Primary Business Requirements:

BRD-AUTH-001

Manage Authorities

BRD-AUTH-002

Validate Authorities

BRD-AUTH-003

Track Authority Lineage

BRD-AUTH-004

Support Authority Evaluation

BRD-AUTH-005

Support Governance Decisions

11.5 Authority Functional Requirements

Mapped Requirements:

SRS-FR-AUTH-001

Create Authority

SRS-FR-AUTH-002

Update Authority

SRS-FR-AUTH-003

Delete Authority

SRS-FR-AUTH-004

Search Authorities

SRS-FR-AUTH-005

View Authority History

11.6 Authority Architecture Traceability

Architecture Component:

SAD-COMP-AUTH

Authority Service

Responsibilities:

Authority Management

Authority Validation

Authority Lineage

Authority Governance

Authority Auditability

11.7 Authority Design Traceability

Technical Design:

TDD-AUTH

Authority Design Module

Includes:

Domain Models

Validation Rules

Credibility Models

Lineage Models

Persistence Logic

11.8 Authority Database Traceability

Database Entity:

DB-AUTHORITY

Authority Table

Stores:

Authority Metadata

Authority Type

Credibility Scores

Ownership

Governance State

Audit References

11.9 Authority API Traceability

API Group:

API-AUTH

Authority APIs

Includes:

POST /authorities

GET /authorities

GET /authorities/{id}

PUT /authorities/{id}

DELETE /authorities/{id}

11.10 Authority Component Traceability

Component:

COMP-AUTHORITY-001

Authority Component

Maps To:

Authority Service

Authority APIs

Authority Repository

Authority Validators

Authority Governance Controls

11.11 Authority Repository Traceability

Repository Layer:

REPO-AUTH

Authority Repository

Responsibilities:

Persistence

Retrieval

Versioning

Lineage Tracking

Audit Linking

11.12 Authority Validation Traceability

Validation Rules:

VAL-AUTH-001

Required Fields

VAL-AUTH-002

Authority Type Validation

VAL-AUTH-003

Credibility Validation

VAL-AUTH-004

Governance Validation

11.13 Authority Lifecycle Traceability

Lifecycle States:

Draft

↓

Review

↓

Verified

↓

Approved

↓

Active

↓

Archived

Each state transition must be traceable.

11.14 Authority Governance Traceability

Governance Controls:

GOV-AUTH-001

Authority Approval

GOV-AUTH-002

Authority Validation

GOV-AUTH-003

Compliance Validation

GOV-AUTH-004

Authority Governance Review

11.15 Authority Audit Traceability

Audit Controls:

AUD-AUTH-001

Creation Audit

AUD-AUTH-002

Modification Audit

AUD-AUTH-003

Deletion Audit

AUD-AUTH-004

Governance Audit

AUD-AUTH-005

Lineage Audit

11.16 Authority Lineage Traceability

Lineage Controls:

LIN-AUTH-001

Source Tracking

LIN-AUTH-002

Authority Relationships

LIN-AUTH-003

Historical Tracking

LIN-AUTH-004

Verification History

11.17 Authority Testing Traceability

Verification Artifacts:

TEST-AUTH-UNIT

Unit Tests

TEST-AUTH-INT

Integration Tests

TEST-AUTH-API

API Tests

TEST-AUTH-SEC

Security Tests

TEST-AUTH-UAT

UAT Tests

11.18 Authority Security Traceability

Security Controls:

SEC-AUTH-001

Authentication

SEC-AUTH-002

Authorization

SEC-AUTH-003

Ownership Controls

SEC-AUTH-004

Audit Logging

11.19 Authority Data Governance Traceability

Data Governance Controls:

DATA-AUTH-001

Ownership Rules

DATA-AUTH-002

Retention Rules

DATA-AUTH-003

Classification Rules

DATA-AUTH-004

Access Controls

11.20 Authority Reporting Traceability

Reports Supported:

Authority Summary Report

Authority Validation Report

Authority Lineage Report

Governance Report

Audit Report

11.21 Authority Release Traceability

Release Controls:

REL-AUTH-001

Functional Validation

REL-AUTH-002

Security Validation

REL-AUTH-003

Governance Validation

REL-AUTH-004

Release Approval

11.22 Authority Coverage Requirements

Every authority capability shall map to:

Requirement

Architecture

Design

Database

API

Testing

Governance

Audit

Release

11.23 Authority Gap Analysis

Authority gaps exist when:

Missing Requirement Mapping

Missing API Mapping

Missing Database Mapping

Missing Test Coverage

Missing Governance Validation

Missing Release Validation

All gaps require remediation.

11.24 Authority Impact Analysis

Changes to authority requirements shall trigger review of:

Architecture

Design

Database

APIs

Tests

Governance Controls

Audit Controls

Lineage Controls

Release Plans

11.25 Coverage Metrics

Required Metrics:

Authority Requirement Coverage %

Authority Test Coverage %

Authority Governance Coverage %

Authority Audit Coverage %

Authority Release Coverage %

Target:

100%

11.26 Governance Rules

RTM-AUTH-GOV-01

Every authority capability must be traceable.

RTM-AUTH-GOV-02

Every authority capability must be tested.

RTM-AUTH-GOV-03

Every authority capability must support governance validation.

RTM-AUTH-GOV-04

Every authority capability must support audits.

RTM-AUTH-GOV-05

Coverage must be measurable.

RTM-AUTH-GOV-06

Release validation is mandatory.

11.27 Architectural Constraints

AC-RTM-AUTH-01

Authority features without requirements are prohibited.

AC-RTM-AUTH-02

Authority features without testing are prohibited.

AC-RTM-AUTH-03

Authority features without governance validation are prohibited.

AC-RTM-AUTH-04

Coverage gaps require remediation.

AC-RTM-AUTH-05

Traceability evidence cannot be deleted.

AC-RTM-AUTH-06

Release approval requires complete authority traceability.

11.28 Section Summary

This section defines the Authority Module Traceability Matrix and establishes mappings between authority requirements, architecture components, technical designs, APIs, database entities, verification artifacts, governance controls, audit controls, lineage controls, and release approvals.

The Authority Module Traceability Matrix ensures that authority management within TODP remains transparent, governed, auditable, lineage-aware, testable, and release-ready throughout its lifecycle.

RTM V1 Progress

❏ Section 1 — Document Overview & Traceability Scope

❏ Section 2 — Traceability Framework

❏ Section 3 — Requirement Classification Model

❏ Section 4 — BRD Traceability Matrix

❏ Section 5 — SRS Traceability Matrix

❏ Section 6 — Architecture Traceability Matrix

❏ Section 7 — Component Traceability Matrix

❏ Section 8 — Database Traceability Matrix

❏ Section 9 — API Traceability Matrix

❏ Section 10 — Opportunity Module Traceability

❏ Section 11 — Authority Module Traceability

❏ Section 12 — Evidence Module Traceability

RTM V1 — Section 12

Evidence Module Traceability

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

12

Version:

V1

Status:

APPROVED

12.1 Purpose

This section defines the Evidence Module Traceability Matrix for TODP.

The Evidence Module manages evidence creation, validation, lineage, relationships, provenance, governance, auditability, and replayability.

Evidence is a foundational decision-support asset within TODP and serves as the primary justification layer for ranking, recommendation, governance, and audit activities.

The objective is to ensure:

- Evidence accountability
- Evidence provenance
- Evidence lineage
- Evidence governance
- Evidence auditability
- Release readiness

12.2 Evidence Traceability Objectives

RTM-EVD-01

Verify evidence requirement coverage.

RTM-EVD-02

Verify evidence implementation traceability.

RTM-EVD-03

Verify evidence governance compliance.

RTM-EVD-04

Verify evidence auditability.

RTM-EVD-05

Verify evidence lineage traceability.

RTM-EVD-06

Verify release readiness.

12.3 Evidence Traceability Model

Business Requirement



System Requirement



Architecture Component



Technical Design



API Contract



Database Entity



Verification Artifact



Governance Control



Audit Artifact



Release Artifact

12.4 Evidence Business Requirements

Primary Business Requirements:

BRD-EVD-001

Manage Evidence

BRD-EVD-002

Track Evidence Relationships

BRD-EVD-003

Maintain Evidence Provenance

BRD-EVD-004

Support Evidence Validation

BRD-EVD-005

Support Governance Decisions

12.5 Evidence Functional Requirements

Mapped Requirements:

SRS-FR-EVD-001

Create Evidence

SRS-FR-EVD-002

Update Evidence

SRS-FR-EVD-003

Delete Evidence

SRS-FR-EVD-004

Search Evidence

SRS-FR-EVD-005

View Evidence History

12.6 Evidence Architecture Traceability

Architecture Component:

SAD-COMP-EVD

Evidence Service

Responsibilities:

Evidence Management

Evidence Validation

Evidence Provenance

Evidence Lineage

Evidence Governance

Evidence Auditability

12.7 Evidence Design Traceability

Technical Design:

TDD-EVD

Evidence Design Module

Includes:

Domain Models

Validation Rules

Relationship Models

Lineage Models

Persistence Logic

12.8 Evidence Database Traceability

Database Entities:

DB-EVIDENCE

Evidence Table

DB-EVIDENCE_LINK

Evidence Relationship Table

Stores:

Evidence Metadata

Evidence Relationships

Evidence Provenance

Governance State

Audit References

Version Information

12.9 Evidence API Traceability

API Group:

API-EVD

Evidence APIs

Includes:

POST /evidence

GET /evidence

GET /evidence/{id}

PUT /evidence/{id}

DELETE /evidence/{id}

GET /evidence/{id}/relationships

12.10 Evidence Component Traceability

Component:

COMP-EVIDENCE-001

Evidence Component

Maps To:

Evidence Service

Evidence APIs

Evidence Repository

Evidence Validators

Evidence Governance Controls

12.11 Evidence Repository Traceability

Repository Layer:

REPO-EVD

Evidence Repository

Responsibilities:

Persistence

Retrieval

Versioning

Relationship Tracking

Lineage Tracking

Audit Linking

12.12 Evidence Validation Traceability

Validation Rules:

VAL-EVD-001

Required Fields

VAL-EVD-002

Evidence Type Validation

VAL-EVD-003

Relationship Validation

VAL-EVD-004

Provenance Validation

VAL-EVD-005

Governance Validation

12.13 Evidence Lifecycle Traceability

Lifecycle States:

Draft

↓

Review

↓

Validated

↓

Approved

↓

Active



Archived

Each state transition must be traceable.

12.14 Evidence Provenance Traceability

Provenance Controls:

PROV-EVD-001

Source Identification

PROV-EVD-002

Origin Tracking

PROV-EVD-003

Source Verification

PROV-EVD-004

Historical Tracking

12.15 Evidence Lineage Traceability

Lineage Controls:

LIN-EVD-001

Parent Relationships

LIN-EVD-002

Child Relationships

LIN-EVD-003

Dependency Tracking

LIN-EVD-004

Relationship History

12.16 Evidence Governance Traceability

Governance Controls:

GOV-EVD-001

Evidence Approval

GOV-EVD-002

Evidence Validation

GOV-EVD-003

Compliance Validation

GOV-EVD-004

Evidence Governance Review

12.17 Evidence Audit Traceability

Audit Controls:

AUD-EVD-001

Creation Audit

AUD-EVD-002

Modification Audit

AUD-EVD-003

Deletion Audit

AUD-EVD-004

Relationship Audit

AUD-EVD-005

Governance Audit

12.18 Evidence Replay Traceability

Replay Controls:

REP-EVD-001

Historical Replay

REP-EVD-002

Relationship Replay

REP-EVD-003

State Reconstruction

REP-EVD-004

Governance Replay

12.19 Evidence Testing Traceability

Verification Artifacts:

TEST-EVD-UNIT

Unit Tests

TEST-EVD-INT

Integration Tests

TEST-EVD-API

API Tests

TEST-EVD-SEC

Security Tests

TEST-EVD-UAT

UAT Tests

12.20 Evidence Security Traceability

Security Controls:

SEC-EVD-001

Authentication

SEC-EVD-002

Authorization

SEC-EVD-003

Ownership Controls

SEC-EVD-004

Audit Logging

SEC-EVD-005

Integrity Protection

12.21 Evidence Data Governance Traceability

Data Governance Controls:

DATA-EVD-001

Ownership Rules

DATA-EVD-002

Retention Rules

DATA-EVD-003

Classification Rules

DATA-EVD-004

Access Controls

DATA-EVD-005

Integrity Controls

12.22 Evidence Reporting Traceability

Reports Supported:

Evidence Summary Report

Evidence Lineage Report

Evidence Provenance Report

Governance Report

Audit Report

Compliance Report

12.23 Evidence Release Traceability

Release Controls:

REL-EVD-001

Functional Validation

REL-EVD-002

Security Validation

REL-EVD-003

Governance Validation

REL-EVD-004

Release Approval

12.24 Evidence Coverage Requirements

Every evidence capability shall map to:

Requirement

Architecture

Design

Database

API

Testing

Governance

Audit

Replay

Release

12.25 Evidence Gap Analysis

Evidence gaps exist when:

Missing Requirement Mapping

Missing API Mapping

Missing Database Mapping

Missing Test Coverage

Missing Governance Validation

Missing Audit Validation

Missing Release Validation

All gaps require remediation.

12.26 Evidence Impact Analysis

Changes to evidence requirements shall trigger review of:

Architecture

Design

Database

Relationships

APIs

Tests

Governance Controls

Audit Controls

Replay Controls

Release Plans

12.27 Coverage Metrics

Required Metrics:

Evidence Requirement Coverage %

Evidence Test Coverage %

Evidence Governance Coverage %

Evidence Audit Coverage %

Evidence Release Coverage %

Target:

100%

12.28 Governance Rules

RTM-EVD-GOV-01

Every evidence capability must be traceable.

RTM-EVD-GOV-02

Every evidence capability must be tested.

RTM-EVD-GOV-03

Every evidence capability must support governance validation.

RTM-EVD-GOV-04

Every evidence capability must support audits.

RTM-EVD-GOV-05

Coverage must be measurable.

RTM-EVD-GOV-06

Release validation is mandatory.

12.29 Architectural Constraints

AC-RTM-EVD-01

Evidence features without requirements are prohibited.

AC-RTM-EVD-02

Evidence features without testing are prohibited.

AC-RTM-EVD-03

Evidence features without governance validation are prohibited.

AC-RTM-EVD-04

Coverage gaps require remediation.

AC-RTM-EVD-05

Traceability evidence cannot be deleted.

AC-RTM-EVD-06

Release approval requires complete evidence traceability.

12.30 Section Summary

This section defines the Evidence Module Traceability Matrix and establishes mappings between evidence requirements, architecture components, technical designs, APIs, database entities, verification artifacts, governance controls, audit controls, provenance controls, lineage controls, replay controls, and release approvals.

The Evidence Module Traceability Matrix ensures that evidence management within TODP remains transparent, lineage-aware, provenance-aware, governed, auditable, replayable, testable, and release-ready throughout its lifecycle.

RTM V1 Progress

- 🔗 Section 1 — Document Overview & Traceability Scope
- 🔗 Section 2 — Traceability Framework
- 🔗 Section 3 — Requirement Classification Model
- 🔗 Section 4 — BRD Traceability Matrix
- 🔗 Section 5 — SRS Traceability Matrix
- 🔗 Section 6 — Architecture Traceability Matrix
- 🔗 Section 7 — Component Traceability Matrix
- 🔗 Section 8 — Database Traceability Matrix
- 🔗 Section 9 — API Traceability Matrix
- 🔗 Section 10 — Opportunity Module Traceability
- 🔗 Section 11 — Authority Module Traceability
- 🔗 Section 12 — Evidence Module Traceability

❏ Section 13 — Retrieval Module Traceability

RTM V1 — Section 13

Retrieval Module Traceability

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

13

Version:

V1

Status:

APPROVED

13.1 Purpose

This section defines the Retrieval Module Traceability Matrix for TODP.

The Retrieval Module is responsible for locating, filtering, ranking candidate evidence and opportunity-related information using keyword search, semantic search, and hybrid retrieval techniques.

The Retrieval Module serves as the foundation for ranking, recommendation, governance review, audit analysis, and replay operations.

The objective is to ensure:

- Retrieval transparency
- Retrieval explainability
- Retrieval accountability
- Retrieval governance
- Retrieval auditability
- Release readiness

13.2 Retrieval Traceability Objectives

RTM-RET-01

Verify retrieval requirement coverage.

RTM-RET-02

Verify retrieval implementation traceability.

RTM-RET-03

Verify retrieval governance compliance.

RTM-RET-04

Verify retrieval auditability.

RTM-RET-05

Verify retrieval replayability.

RTM-RET-06

Verify release readiness.

13.3 Retrieval Traceability Model

Business Requirement



System Requirement



Architecture Component



Technical Design



Search Infrastructure



API Contract



Database Entity



Verification Artifact



Governance Control



Audit Artifact



Release Artifact

13.4 Retrieval Business Requirements

Primary Business Requirements:

BRD-RET-001

Search Information

BRD-RET-002

Support Semantic Retrieval

BRD-RET-003

Support Hybrid Retrieval

BRD-RET-004

Support Explainable Search

BRD-RET-005

Support Governance Review

13.5 Retrieval Functional Requirements

Mapped Requirements:

SRS-FR-RET-001

Keyword Search

SRS-FR-RET-002

Semantic Search

SRS-FR-RET-003

Hybrid Search

SRS-FR-RET-004

Search Filtering

SRS-FR-RET-005

Search History

13.6 Retrieval Architecture Traceability

Architecture Component:

SAD-COMP-RET

Retrieval Engine

Responsibilities:

Query Processing

Hybrid Retrieval

Candidate Selection

Result Aggregation

Search Explainability

Governance Support

13.7 Retrieval Design Traceability

Technical Design:

TDD-RET

Retrieval Design Module

Includes:

Search Models

Query Processing

Vector Retrieval

BM25 Retrieval

Hybrid Scoring

Ranking Preparation

13.8 Retrieval Database Traceability

Database Entities:

DB-RETRIEVAL_HISTORY

Retrieval History Table

DB-EMBEDDING

Embedding Store

Stores:

Search Requests

Search Responses

Embedding References

Search Metadata

Governance References

Audit References

13.9 Retrieval API Traceability

API Group:

API-RET

Retrieval APIs

Includes:

POST /retrieve

POST /search

POST /search/hybrid

GET /search/history

GET /search/{id}

13.10 Retrieval Component Traceability

Component:

COMP-RET-001

Retrieval Component

Maps To:

Retrieval Engine

Search APIs

Embedding Services

Search Repositories

Governance Controls

13.11 Retrieval Repository Traceability

Repository Layer:

REPO-RET

Retrieval Repository

Responsibilities:

Search Storage

Search Retrieval

Embedding Access

History Tracking

Audit Linking

13.12 Query Processing Traceability

Processing Controls:

RET-QRY-001

Query Validation

RET-QRY-002

Query Parsing

RET-QRY-003

Query Expansion

RET-QRY-004

Query Normalization

13.13 Keyword Search Traceability

Search Controls:

RET-KW-001

Keyword Search

RET-KW-002

Keyword Matching

RET-KW-003

Keyword Filtering

RET-KW-004

Keyword Explainability

13.14 Semantic Search Traceability

Search Controls:

RET-SEM-001

Embedding Search

RET-SEM-002

Vector Similarity

RET-SEM-003

Semantic Filtering

RET-SEM-004

Semantic Explainability

13.15 Hybrid Search Traceability

Search Controls:

RET-HYB-001

Hybrid Retrieval

RET-HYB-002

BM25 Integration

RET-HYB-003

Vector Integration

RET-HYB-004

Hybrid Scoring

RET-HYB-005

Hybrid Explainability

13.16 Search Explainability Traceability

Explainability Controls:

EXP-RET-001

Match Explanation

EXP-RET-002

Score Explanation

EXP-RET-003

Ranking Explanation

EXP-RET-004

Result Explanation

13.17 Retrieval Governance Traceability

Governance Controls:

GOV-RET-001

Search Governance

GOV-RET-002

Search Approval Rules

GOV-RET-003

Compliance Validation

GOV-RET-004

Retrieval Governance Review

13.18 Retrieval Audit Traceability

Audit Controls:

AUD-RET-001

Search Audit

AUD-RET-002

Query Audit

AUD-RET-003

Result Audit

AUD-RET-004

Governance Audit

AUD-RET-005

Replay Audit

13.19 Retrieval Replay Traceability

Replay Controls:

REP-RET-001

Historical Search Replay

REP-RET-002

Query Reconstruction

REP-RET-003

Result Reconstruction

REP-RET-004

Governance Replay

13.20 Retrieval Testing Traceability

Verification Artifacts:

TEST-RET-UNIT

Unit Tests

TEST-RET-INT

Integration Tests

TEST-RET-API

API Tests

TEST-RET-PERF

Performance Tests

TEST-RET-UAT

UAT Tests

13.21 Retrieval Security Traceability

Security Controls:

SEC-RET-001

Authentication

SEC-RET-002

Authorization

SEC-RET-003

Query Protection

SEC-RET-004

Audit Logging

SEC-RET-005

Access Controls

13.22 Retrieval Data Governance Traceability

Data Governance Controls:

DATA-RET-001

Ownership Rules

DATA-RET-002

Retention Rules

DATA-RET-003

Search History Controls

DATA-RET-004

Access Controls

DATA-RET-005

Compliance Controls

13.23 Retrieval Reporting Traceability

Reports Supported:

Search Activity Report

Search Performance Report

Search Explainability Report

Governance Report

Audit Report

Replay Report

13.24 Retrieval Release Traceability

Release Controls:

REL-RET-001

Functional Validation

REL-RET-002

Performance Validation

REL-RET-003

Governance Validation

REL-RET-004

Release Approval

13.25 Retrieval Coverage Requirements

Every retrieval capability shall map to:

Requirement

Architecture

Design

Infrastructure

Database

API

Testing

Governance

Audit

Replay

Release

13.26 Retrieval Gap Analysis

Retrieval gaps exist when:

Missing Requirement Mapping

Missing Search Control

Missing API Mapping

Missing Test Coverage

Missing Governance Validation

Missing Audit Validation

Missing Release Validation

All gaps require remediation.

13.27 Retrieval Impact Analysis

Changes to retrieval requirements shall trigger review of:

Architecture

Design

Embeddings

Search Infrastructure

APIs

Tests

Governance Controls

Audit Controls

Replay Controls

Release Plans

13.28 Coverage Metrics

Required Metrics:

Retrieval Requirement Coverage %

Retrieval Test Coverage %

Retrieval Governance Coverage %

Retrieval Audit Coverage %

Retrieval Release Coverage %

Target:

100%

13.29 Governance Rules

RTM-RET-GOV-01

Every retrieval capability must be traceable.

RTM-RET-GOV-02

Every retrieval capability must be tested.

RTM-RET-GOV-03

Every retrieval capability must support governance validation.

RTM-RET-GOV-04

Every retrieval capability must support audits.

RTM-RET-GOV-05

Coverage must be measurable.

RTM-RET-GOV-06

Release validation is mandatory.

13.30 Architectural Constraints

AC-RTM-RET-01

Retrieval features without requirements are prohibited.

AC-RTM-RET-02

Retrieval features without testing are prohibited.

AC-RTM-RET-03

Retrieval features without governance validation are prohibited.

AC-RTM-RET-04

Coverage gaps require remediation.

AC-RTM-RET-05

Traceability evidence cannot be deleted.

AC-RTM-RET-06

Release approval requires complete retrieval traceability.

13.31 Section Summary

This section defines the Retrieval Module Traceability Matrix and establishes mappings between retrieval requirements, architecture components, technical designs, search infrastructure, APIs, database entities, verification artifacts, governance controls, audit controls, replay controls, and release approvals.

The Retrieval Module Traceability Matrix ensures that retrieval operations within TODP remain transparent, explainable, governed, auditable, replayable, testable, and release-ready throughout their lifecycle.

RTM V1 Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Framework

☐ Section 3 — Requirement Classification Model

☐ Section 4 — BRD Traceability Matrix

☐ Section 5 — SRS Traceability Matrix

☐ Section 6 — Architecture Traceability Matrix

☐ Section 7 — Component Traceability Matrix

☐ Section 8 — Database Traceability Matrix

☐ Section 9 — API Traceability Matrix

☐ Section 10 — Opportunity Module Traceability

☐ Section 11 — Authority Module Traceability

☐ Section 12 — Evidence Module Traceability

☐ Section 13 — Retrieval Module Traceability

☐ Section 14 — Ranking Module Traceability

RTM V1 — Section 14

Ranking Module Traceability

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

14

Version:

V1

Status:

APPROVED

14.1 Purpose

This section defines the Ranking Module Traceability Matrix for TODP.

The Ranking Module is responsible for evaluating retrieved candidates and calculating transparent, explainable, auditable, and reproducible ranking scores.

The Ranking Module serves as the decision-prioritization layer between retrieval and recommendation generation.

The objective is to ensure:

- Ranking transparency
- Ranking explainability
- Ranking accountability
- Ranking governance
- Ranking auditability

- Release readiness

14.2 Ranking Traceability Objectives

RTM-RANK-01

Verify ranking requirement coverage.

RTM-RANK-02

Verify ranking implementation traceability.

RTM-RANK-03

Verify ranking governance compliance.

RTM-RANK-04

Verify ranking auditability.

RTM-RANK-05

Verify ranking replayability.

RTM-RANK-06

Verify release readiness.

14.3 Ranking Traceability Model

Business Requirement



System Requirement



Architecture Component



Technical Design



Ranking Logic



API Contract



Database Entity



Verification Artifact



Governance Control



Audit Artifact



Release Artifact

14.4 Ranking Business Requirements

Primary Business Requirements:

BRD-RANK-001

Rank Opportunities

BRD-RANK-002

Support Transparent Scoring

BRD-RANK-003

Support Explainable Ranking

BRD-RANK-004

Support Governance Review

BRD-RANK-005

Support Replay Validation

14.5 Ranking Functional Requirements

Mapped Requirements:

SRS-FR-RANK-001

Calculate Rankings

SRS-FR-RANK-002

Apply Scoring Models

SRS-FR-RANK-003

Apply Ranking Rules

SRS-FR-RANK-004

Provide Ranking Explanations

SRS-FR-RANK-005

Store Ranking History

14.6 Ranking Architecture Traceability

Architecture Component:

SAD-COMP-RANK

Ranking Engine

Responsibilities:

Score Calculation

Candidate Evaluation

Weight Application

Ranking Explainability

Governance Support

Audit Support

14.7 Ranking Design Traceability

Technical Design:

TDD-RANK

Ranking Design Module

Includes:

Scoring Models

Weight Models

Ranking Rules

Evaluation Logic

Ranking Persistence

14.8 Ranking Database Traceability

Database Entity:

DB-RANKING

Ranking Table

Stores:

Ranking Scores

Weight Values

Ranking Metadata

Explanation Metadata

Governance References

Audit References

14.9 Ranking API Traceability

API Group:

API-RANK

Ranking APIs

Includes:

POST /rank

GET /rankings

GET /rankings/{id}

GET /rankings/{id}/explanation

GET /rankings/history

14.10 Ranking Component Traceability

Component:

COMP-RANK-001

Ranking Component

Maps To:

Ranking Engine

Ranking APIs

Ranking Repository

Ranking Validators

Governance Controls

14.11 Ranking Repository Traceability

Repository Layer:

REPO-RANK

Ranking Repository

Responsibilities:

Persistence

Retrieval

Versioning

History Tracking

Audit Linking

14.12 Scoring Model Traceability

Scoring Controls:

RANK-SCORE-001

Score Calculation

RANK-SCORE-002

Weight Application

RANK-SCORE-003

Normalization

RANK-SCORE-004

Threshold Evaluation

14.13 Ranking Rule Traceability

Rule Controls:

RANK-RULE-001

Business Rules

RANK-RULE-002

Governance Rules

RANK-RULE-003

Priority Rules

RANK-RULE-004

Decision Rules

14.14 Ranking Explainability Traceability

Explainability Controls:

EXP-RANK-001

Score Explanation

EXP-RANK-002

Weight Explanation

EXP-RANK-003

Rule Explanation

EXP-RANK-004

Decision Explanation

14.15 Ranking Governance Traceability

Governance Controls:

GOV-RANK-001

Ranking Governance

GOV-RANK-002

Ranking Approval Rules

GOV-RANK-003

Compliance Validation

GOV-RANK-004

Ranking Governance Review

14.16 Ranking Audit Traceability

Audit Controls:

AUD-RANK-001

Ranking Audit

AUD-RANK-002

Score Audit

AUD-RANK-003

Rule Audit

AUD-RANK-004

Governance Audit

AUD-RANK-005

Replay Audit

14.17 Ranking Replay Traceability

Replay Controls:

REP-RANK-001

Historical Ranking Replay

REP-RANK-002

Score Reconstruction

REP-RANK-003

Rule Reconstruction

REP-RANK-004

Governance Replay

14.18 Ranking Testing Traceability

Verification Artifacts:

TEST-RANK-UNIT

Unit Tests

TEST-RANK-INT

Integration Tests

TEST-RANK-API

API Tests

TEST-RANK-PERF

Performance Tests

TEST-RANK-UAT

UAT Tests

14.19 Ranking Security Traceability

Security Controls:

SEC-RANK-001

Authentication

SEC-RANK-002

Authorization

SEC-RANK-003

Access Controls

SEC-RANK-004

Audit Logging

SEC-RANK-005

Integrity Protection

14.20 Ranking Data Governance Traceability

Data Governance Controls:

DATA-RANK-001

Ownership Rules

DATA-RANK-002

Retention Rules

DATA-RANK-003

Classification Rules

DATA-RANK-004

Access Controls

DATA-RANK-005

Compliance Controls

14.21 Ranking Reporting Traceability

Reports Supported:

Ranking Summary Report

Ranking Explanation Report

Score Analysis Report

Governance Report

Audit Report

Replay Report

14.22 Ranking Release Traceability

Release Controls:

REL-RANK-001

Functional Validation

REL-RANK-002

Performance Validation

REL-RANK-003

Governance Validation

REL-RANK-004

Release Approval

14.23 Ranking Coverage Requirements

Every ranking capability shall map to:

Requirement

Architecture

Design

Ranking Logic

Database

API

Testing

Governance

Audit

Replay

Release

14.24 Ranking Gap Analysis

Ranking gaps exist when:

Missing Requirement Mapping

Missing Scoring Logic

Missing API Mapping

Missing Test Coverage

Missing Governance Validation

Missing Audit Validation

Missing Release Validation

All gaps require remediation.

14.25 Ranking Impact Analysis

Changes to ranking requirements shall trigger review of:

Architecture

Design

Scoring Models

Ranking Rules

APIs

Tests

Governance Controls

Audit Controls

Replay Controls

Release Plans

14.26 Coverage Metrics

Required Metrics:

Ranking Requirement Coverage %

Ranking Test Coverage %

Ranking Governance Coverage %

Ranking Audit Coverage %

Ranking Release Coverage %

Target:

100%

14.27 Governance Rules

RTM-RANK-GOV-01

Every ranking capability must be traceable.

RTM-RANK-GOV-02

Every ranking capability must be tested.

RTM-RANK-GOV-03

Every ranking capability must support governance validation.

RTM-RANK-GOV-04

Every ranking capability must support audits.

RTM-RANK-GOV-05

Coverage must be measurable.

RTM-RANK-GOV-06

Release validation is mandatory.

14.28 Architectural Constraints

AC-RTM-RANK-01

Ranking features without requirements are prohibited.

AC-RTM-RANK-02

Ranking features without testing are prohibited.

AC-RTM-RANK-03

Ranking features without governance validation are prohibited.

AC-RTM-RANK-04

Coverage gaps require remediation.

AC-RTM-RANK-05

Traceability evidence cannot be deleted.

AC-RTM-RANK-06

Release approval requires complete ranking traceability.

14.29 Section Summary

This section defines the Ranking Module Traceability Matrix and establishes mappings between ranking requirements, architecture components, technical designs, scoring models, APIs, database entities, verification artifacts, governance controls, audit controls, replay controls, and release approvals.

The Ranking Module Traceability Matrix ensures that ranking operations within TODP remain transparent, explainable, governed, auditable, replayable, testable, and release-ready throughout their lifecycle.

RTM V1 Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Framework

☐ Section 3 — Requirement Classification Model

☐ Section 4 — BRD Traceability Matrix

☐ Section 5 — SRS Traceability Matrix

☐ Section 6 — Architecture Traceability Matrix

☐ Section 7 — Component Traceability Matrix

☐ Section 8 — Database Traceability Matrix

☐ Section 9 — API Traceability Matrix

☐ Section 10 — Opportunity Module Traceability

☐ Section 11 — Authority Module Traceability

☐ Section 12 — Evidence Module Traceability

☐ Section 13 — Retrieval Module Traceability

☐ Section 14 — Ranking Module Traceability

☐ Section 15 — Recommendation Module Traceability

RTM V1 — Section 15

Recommendation Module Traceability

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

15

Version:

V1

Status:

APPROVED

15.1 Purpose

This section defines the Recommendation Module Traceability Matrix for TODP.

The Recommendation Module is responsible for generating transparent, explainable, auditable, governance-compliant recommendations based on retrieved evidence, authority signals, ranking outputs, policies, and business objectives.

The Recommendation Module represents the final decision-support layer before governance approval and operational action.

The objective is to ensure:

- Recommendation transparency
- Recommendation explainability
- Recommendation accountability
- Recommendation governance
- Recommendation auditability

- Release readiness

15.2 Recommendation Traceability Objectives

RTM-REC-01

Verify recommendation requirement coverage.

RTM-REC-02

Verify recommendation implementation traceability.

RTM-REC-03

Verify recommendation governance compliance.

RTM-REC-04

Verify recommendation auditability.

RTM-REC-05

Verify recommendation replayability.

RTM-REC-06

Verify release readiness.

15.3 Recommendation Traceability Model

Business Requirement



System Requirement



Architecture Component



Technical Design



Recommendation Logic



API Contract



Database Entity



Verification Artifact



Governance Control



Audit Artifact



Release Artifact

15.4 Recommendation Business Requirements

Primary Business Requirements:

BRD-REC-001

Generate Recommendations

BRD-REC-002

Provide Recommendation Explanations

BRD-REC-003

Support Governance Review

BRD-REC-004

Support Decision Transparency

BRD-REC-005

Support Recommendation Replay

15.5 Recommendation Functional Requirements

Mapped Requirements:

SRS-FR-REC-001

Generate Recommendations

SRS-FR-REC-002

Generate Explanations

SRS-FR-REC-003

Store Recommendation History

SRS-FR-REC-004

Support Recommendation Review

SRS-FR-REC-005

Support Recommendation Replay

15.6 Recommendation Architecture Traceability

Architecture Component:

SAD-COMP-REC

Recommendation Engine

Responsibilities:

Recommendation Generation

Decision Logic

Explanation Generation

Governance Support

Audit Support

Replay Support

15.7 Recommendation Design Traceability

Technical Design:

TDD-REC

Recommendation Design Module

Includes:

Decision Models

Recommendation Rules

Explanation Models

Confidence Models

Persistence Logic

15.8 Recommendation Database Traceability

Database Entity:

DB-RECOMMENDATION

Recommendation Table

Stores:

Recommendation Outputs

Confidence Scores

Explanation Metadata

Governance References

Audit References

Replay References

15.9 Recommendation API Traceability

API Group:

API-REC

Recommendation APIs

Includes:

POST /recommend

GET /recommendations

GET /recommendations/{id}

GET /recommendations/{id}/explanation

GET /recommendations/history

15.10 Recommendation Component Traceability

Component:

COMP-REC-001

Recommendation Component

Maps To:

Recommendation Engine

Recommendation APIs

Recommendation Repository

Recommendation Validators

Governance Controls

15.11 Recommendation Repository Traceability

Repository Layer:

REPO-REC

Recommendation Repository

Responsibilities:

Persistence

Retrieval

Versioning

History Tracking

Audit Linking

Replay Linking

15.12 Recommendation Rule Traceability

Recommendation Controls:

REC-RULE-001

Recommendation Rules

REC-RULE-002

Business Logic

REC-RULE-003

Priority Logic

REC-RULE-004

Decision Logic

15.13 Recommendation Confidence Traceability

Confidence Controls:

REC-CONF-001

Confidence Calculation

REC-CONF-002

Confidence Thresholds

REC-CONF-003

Confidence Validation

REC-CONF-004

Confidence Reporting

15.14 Recommendation Explainability Traceability

Explainability Controls:

EXP-REC-001

Recommendation Explanation

EXP-REC-002

Evidence Explanation

EXP-REC-003

Authority Explanation

EXP-REC-004

Ranking Explanation

EXP-REC-005

Decision Explanation

15.15 Recommendation Governance Traceability

Governance Controls:

GOV-REC-001

Recommendation Governance

GOV-REC-002

Approval Rules

GOV-REC-003

Compliance Validation

GOV-REC-004

Governance Review

15.16 Recommendation Audit Traceability

Audit Controls:

AUD-REC-001

Recommendation Audit

AUD-REC-002

Decision Audit

AUD-REC-003

Explanation Audit

AUD-REC-004

Governance Audit

AUD-REC-005

Replay Audit

15.17 Recommendation Replay Traceability

Replay Controls:

REP-REC-001

Historical Recommendation Replay

REP-REC-002

Decision Reconstruction

REP-REC-003

Explanation Reconstruction

REP-REC-004

Governance Replay

15.18 Recommendation Testing Traceability

Verification Artifacts:

TEST-REC-UNIT

Unit Tests

TEST-REC-INT

Integration Tests

TEST-REC-API

API Tests

TEST-REC-PERF

Performance Tests

TEST-REC-UAT

UAT Tests

15.19 Recommendation Security Traceability

Security Controls:

SEC-REC-001

Authentication

SEC-REC-002

Authorization

SEC-REC-003

Access Controls

SEC-REC-004

Audit Logging

SEC-REC-005

Integrity Protection

15.20 Recommendation Data Governance Traceability

Data Governance Controls:

DATA-REC-001

Ownership Rules

DATA-REC-002

Retention Rules

DATA-REC-003

Classification Rules

DATA-REC-004

Access Controls

DATA-REC-005

Compliance Controls

15.21 Recommendation Reporting Traceability

Reports Supported:

Recommendation Summary Report

Recommendation Explanation Report

Confidence Analysis Report

Governance Report

Audit Report

Replay Report

15.22 Recommendation Release Traceability

Release Controls:

REL-REC-001

Functional Validation

REL-REC-002

Performance Validation

REL-REC-003

Governance Validation

REL-REC-004

Release Approval

15.23 Recommendation Coverage Requirements

Every recommendation capability shall map to:

Requirement

Architecture

Design

Recommendation Logic

Database

API

Testing

Governance

Audit

Replay

Release

15.24 Recommendation Gap Analysis

Recommendation gaps exist when:

Missing Requirement Mapping

Missing Recommendation Logic

Missing API Mapping

Missing Test Coverage

Missing Governance Validation

Missing Audit Validation

Missing Release Validation

All gaps require remediation.

15.25 Recommendation Impact Analysis

Changes to recommendation requirements shall trigger review of:

Architecture

Design

Recommendation Rules

Confidence Models

Explanation Models

APIs

Tests

Governance Controls

Audit Controls

Replay Controls

Release Plans

15.26 Coverage Metrics

Required Metrics:

Recommendation Requirement Coverage %

Recommendation Test Coverage %

Recommendation Governance Coverage %

Recommendation Audit Coverage %

Recommendation Release Coverage %

Target:

100%

15.27 Governance Rules

RTM-REC-GOV-01

Every recommendation capability must be traceable.

RTM-REC-GOV-02

Every recommendation capability must be tested.

RTM-REC-GOV-03

Every recommendation capability must support governance validation.

RTM-REC-GOV-04

Every recommendation capability must support audits.

RTM-REC-GOV-05

Coverage must be measurable.

RTM-REC-GOV-06

Release validation is mandatory.

15.28 Architectural Constraints

AC-RTM-REC-01

Recommendation features without requirements are prohibited.

AC-RTM-REC-02

Recommendation features without testing are prohibited.

AC-RTM-REC-03

Recommendation features without governance validation are prohibited.

AC-RTM-REC-04

Coverage gaps require remediation.

AC-RTM-REC-05

Traceability evidence cannot be deleted.

AC-RTM-REC-06

Release approval requires complete recommendation traceability.

15.29 Section Summary

This section defines the Recommendation Module Traceability Matrix and establishes mappings between recommendation requirements, architecture components, technical designs, recommendation rules, confidence models, APIs, database entities, verification artifacts, governance controls, audit controls, replay controls, and release approvals.

The Recommendation Module Traceability Matrix ensures that recommendation operations within TODP remain transparent, explainable, governed, auditable, replayable, testable, and release-ready throughout their lifecycle.

RTM V1 Progress

🔗 Section 1 — Document Overview & Traceability Scope

🔗 Section 2 — Traceability Framework

🔗 Section 3 — Requirement Classification Model

🔗 Section 4 — BRD Traceability Matrix

🔗 Section 5 — SRS Traceability Matrix

🔗 Section 6 — Architecture Traceability Matrix

🔗 Section 7 — Component Traceability Matrix

🔗 Section 8 — Database Traceability Matrix

🔗 Section 9 — API Traceability Matrix

🔗 Section 10 — Opportunity Module Traceability

🔗 Section 11 — Authority Module Traceability

🔗 Section 12 — Evidence Module Traceability

🔗 Section 13 — Retrieval Module Traceability

🔗 Section 14 — Ranking Module Traceability

🔗 Section 15 — Recommendation Module Traceability

▣ Section 16 — Governance Traceability

RTM V1 — Section 16

Governance Traceability

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

16

Version:

V1

Status:

APPROVED

16.1 Purpose

This section defines the Governance Traceability Matrix for TODP.

Governance traceability ensures that all business requirements, system requirements, architectural decisions, implementation artifacts, ranking outcomes, recommendations, audits, and releases are governed through approved policies, controls, reviews, and compliance mechanisms.

The objective is to ensure:

- Governance accountability
- Policy enforcement
- Compliance readiness
- Decision transparency
- Audit readiness
- Release authorization

16.2 Governance Traceability Objectives

RTM-GOV-01

Verify governance requirement coverage.

RTM-GOV-02

Verify policy enforcement traceability.

RTM-GOV-03

Verify compliance traceability.

RTM-GOV-04

Verify audit readiness.

RTM-GOV-05

Verify approval traceability.

RTM-GOV-06

Verify release readiness.

16.3 Governance Traceability Model

Business Requirement



System Requirement



Governance Policy



Governance Control



Governance Workflow



Approval Record



Audit Evidence



Release Decision

16.4 Governance Domains

The following governance domains shall be traced:

Business Governance

Operational Governance

Security Governance

Data Governance

Decision Governance

Compliance Governance

Audit Governance

Release Governance

16.5 Governance Business Requirements

Primary Business Requirements:

BRD-GOV-001

Govern Platform Operations

BRD-GOV-002

Enforce Policies

BRD-GOV-003

Support Compliance Reviews

BRD-GOV-004

Support Audit Reviews

BRD-GOV-005

Support Release Decisions

16.6 Governance Functional Requirements

Mapped Requirements:

SRS-FR-GOV-001

Policy Enforcement

SRS-FR-GOV-002

Approval Workflows

SRS-FR-GOV-003

Compliance Validation

SRS-FR-GOV-004

Exception Handling

SRS-FR-GOV-005

Governance Reporting

16.7 Governance Architecture Traceability

Architecture Component:

SAD-COMP-GOV

Governance Service

Responsibilities:

Policy Enforcement

Workflow Execution

Approval Management

Compliance Validation

Governance Reporting

16.8 Governance Design Traceability

Technical Design:

TDD-GOV

Governance Design Module

Includes:

Policy Models

Workflow Models

Approval Models

Exception Models

Compliance Models

16.9 Governance Database Traceability

Database Entity:

DB-GOVERNANCE

Governance Table

Stores:

Policies

Approvals

Governance Actions

Compliance References

Audit References

Release References

16.10 Governance API Traceability

API Group:

API-GOV

Governance APIs

Includes:

POST /governance/policies

POST /governance/reviews

POST /governance/approvals

GET /governance/history

GET /governance/reports

16.11 Governance Component Traceability

Component:

COMP-GOV-001

Governance Component

Maps To:

Governance Service

Governance APIs

Governance Repository

Governance Validators

Compliance Controls

16.12 Policy Traceability

Policy Controls:

POL-GOV-001

Business Policies

POL-GOV-002

Security Policies

POL-GOV-003

Data Policies

POL-GOV-004

Compliance Policies

POL-GOV-005

Release Policies

16.13 Workflow Traceability

Workflow Controls:

WF-GOV-001

Approval Workflow

WF-GOV-002

Review Workflow

WF-GOV-003

Exception Workflow

WF-GOV-004

Escalation Workflow

16.14 Compliance Traceability

Compliance Controls:

COMP-GOV-001

Compliance Validation

COMP-GOV-002

Compliance Reviews

COMP-GOV-003

Compliance Reporting

COMP-GOV-004

Compliance Evidence

16.15 Governance Approval Traceability

Approval Controls:

APR-GOV-001

Policy Approval

APR-GOV-002

Decision Approval

APR-GOV-003

Exception Approval

APR-GOV-004

Release Approval

16.16 Governance Exception Traceability

Exception Controls:

EX-GOV-001

Policy Exceptions

EX-GOV-002

Workflow Exceptions

EX-GOV-003

Compliance Exceptions

EX-GOV-004

Release Exceptions

16.17 Governance Audit Traceability

Audit Controls:

AUD-GOV-001

Policy Audit

AUD-GOV-002

Approval Audit

AUD-GOV-003

Compliance Audit

AUD-GOV-004

Governance Audit

AUD-GOV-005

Release Audit

16.18 Governance Replay Traceability

Replay Controls:

REP-GOV-001

Decision Replay

REP-GOV-002

Approval Replay

REP-GOV-003

Policy Replay

REP-GOV-004

Governance Reconstruction

16.19 Governance Testing Traceability

Verification Artifacts:

TEST-GOV-UNIT

Unit Tests

TEST-GOV-INT

Integration Tests

TEST-GOV-API

API Tests

TEST-GOV-COMP

Compliance Tests

TEST-GOV-UAT

UAT Tests

16.20 Governance Security Traceability

Security Controls:

SEC-GOV-001

Authentication

SEC-GOV-002

Authorization

SEC-GOV-003

Policy Access Controls

SEC-GOV-004

Audit Logging

SEC-GOV-005

Integrity Protection

16.21 Governance Data Governance Traceability

Data Governance Controls:

DATA-GOV-001

Ownership Rules

DATA-GOV-002

Retention Rules

DATA-GOV-003

Classification Rules

DATA-GOV-004

Access Controls

DATA-GOV-005

Compliance Controls

16.22 Governance Reporting Traceability

Reports Supported:

Governance Summary Report

Policy Compliance Report

Approval Report

Exception Report

Audit Report

Release Report

16.23 Governance Release Traceability

Release Controls:

REL-GOV-001

Governance Validation

REL-GOV-002

Compliance Validation

REL-GOV-003

Approval Validation

REL-GOV-004

Release Authorization

16.24 Governance Coverage Requirements

Every governance capability shall map to:

Requirement

Architecture

Design

Policy

Workflow

Database

API

Testing

Audit

Replay

Release

16.25 Governance Gap Analysis

Governance gaps exist when:

Missing Requirement Mapping

Missing Policy Mapping

Missing Workflow Mapping

Missing Test Coverage

Missing Audit Validation

Missing Release Validation

All gaps require remediation.

16.26 Governance Impact Analysis

Changes to governance requirements shall trigger review of:

Policies

Workflows

Approvals

Compliance Controls

APIs

Tests

Audit Controls

Replay Controls

Release Plans

16.27 Coverage Metrics

Required Metrics:

Governance Requirement Coverage %

Governance Test Coverage %

Governance Audit Coverage %

Governance Compliance Coverage %

Governance Release Coverage %

Target:

100%

16.28 Governance Rules

RTM-GOV-GOV-01

Every governance capability must be traceable.

RTM-GOV-GOV-02

Every governance capability must be tested.

RTM-GOV-GOV-03

Every governance capability must support compliance validation.

RTM-GOV-GOV-04

Every governance capability must support audits.

RTM-GOV-GOV-05

Coverage must be measurable.

RTM-GOV-GOV-06

Release authorization is mandatory.

16.29 Architectural Constraints

AC-RTM-GOV-01

Governance features without requirements are prohibited.

AC-RTM-GOV-02

Governance features without testing are prohibited.

AC-RTM-GOV-03

Governance features without compliance validation are prohibited.

AC-RTM-GOV-04

Coverage gaps require remediation.

AC-RTM-GOV-05

Traceability evidence cannot be deleted.

AC-RTM-GOV-06

Release approval requires complete governance traceability.

16.30 Section Summary

This section defines the Governance Traceability Matrix and establishes mappings between governance requirements, policies, workflows, approvals, compliance controls, architecture components, APIs, database entities, verification artifacts, audit controls, replay controls, and release authorizations.

The Governance Traceability Matrix ensures that governance operations within TODP remain transparent, compliant, auditable, reproducible, testable, and release-authorized throughout their lifecycle.

RTM V1 Progress

📄 Section 1 — Document Overview & Traceability Scope

📄 Section 2 — Traceability Framework

📄 Section 3 — Requirement Classification Model

- ❏ Section 4 — BRD Traceability Matrix
- ❏ Section 5 — SRS Traceability Matrix
- ❏ Section 6 — Architecture Traceability Matrix
- ❏ Section 7 — Component Traceability Matrix
- ❏ Section 8 — Database Traceability Matrix
- ❏ Section 9 — API Traceability Matrix
- ❏ Section 10 — Opportunity Module Traceability
- ❏ Section 11 — Authority Module Traceability
- ❏ Section 12 — Evidence Module Traceability
- ❏ Section 13 — Retrieval Module Traceability
- ❏ Section 14 — Ranking Module Traceability
- ❏ Section 15 — Recommendation Module Traceability
- ❏ Section 16 — Governance Traceability
- ❏ Section 17 — Audit & Replay Traceability

RTM V1 — Section 17

Audit & Replay Traceability

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

Version:

V1

Status:

APPROVED

17.1 Purpose

This section defines the Audit & Replay Traceability Matrix for TODP.

Audit and Replay capabilities ensure that every decision, recommendation, ranking outcome, governance action, retrieval result, and system state can be reconstructed, verified, reviewed, and explained.

The Audit & Replay framework provides the foundation for transparency, accountability, governance validation, compliance readiness, and enterprise trust.

The objective is to ensure:

- Complete auditability
- Complete replayability
- Decision transparency
- Governance accountability
- Compliance readiness
- Release confidence

17.2 Audit & Replay Traceability Objectives

RTM-AUD-01

Verify audit coverage.

RTM-AUD-02

Verify replay coverage.

RTM-AUD-03

Verify decision traceability.

RTM-AUD-04

Verify governance traceability.

RTM-AUD-05

Verify compliance readiness.

RTM-AUD-06

Verify release readiness.

17.3 Audit & Replay Traceability Model

Business Requirement



System Requirement



Audit Control



Replay Control



Audit Artifact



Snapshot Artifact



Verification Artifact



Governance Approval



Release Validation

17.4 Audit & Replay Domains

The following domains shall be traced:

Opportunity Audits

Authority Audits

Evidence Audits

Retrieval Audits

Ranking Audits

Recommendation Audits

Governance Audits

Release Audits

Replay Operations

System Reconstruction

17.5 Audit Business Requirements

Primary Business Requirements:

BRD-AUD-001

Support Audits

BRD-AUD-002

Support Decision Transparency

BRD-AUD-003

Support Historical Reviews

BRD-AUD-004

Support Compliance Reviews

BRD-AUD-005

Support Replay Validation

17.6 Audit Functional Requirements

Mapped Requirements:

SRS-FR-AUD-001

Audit Logging

SRS-FR-AUD-002

Audit Search

SRS-FR-AUD-003

Audit Reporting

SRS-FR-AUD-004

Audit Retention

SRS-FR-AUD-005

Audit Replay

17.7 Audit Architecture Traceability

Architecture Component:

SAD-COMP-AUD

Audit Service

Responsibilities:

Audit Logging

Audit Retrieval

Audit Reporting

Compliance Support

Governance Support

Replay Support

17.8 Replay Architecture Traceability

Architecture Component:

SAD-COMP-REP

Replay Engine

Responsibilities:

Historical Reconstruction

Decision Replay

State Replay

Governance Replay

Audit Validation

17.9 Audit Design Traceability

Technical Design:

TDD-AUD

Audit Design Module

Includes:

Audit Models

Retention Models

Search Models

Compliance Models

Reporting Models

17.10 Replay Design Traceability

Technical Design:

TDD-REP

Replay Design Module

Includes:

Replay Models

Snapshot Models

State Models

Reconstruction Models

Validation Models

17.11 Audit Database Traceability

Database Entity:

DB-AUDIT

Audit Table

Stores:

Audit Events

Audit Metadata

User Actions

System Actions

Governance Actions

Release Actions

17.12 Replay Database Traceability

Database Entity:

DB-SNAPSHOT

Snapshot Table

Stores:

System Snapshots

Decision States

Ranking States

Recommendation States

Governance States

Replay References

17.13 Audit API Traceability

API Group:

API-AUD

Audit APIs

Includes:

GET /audit

GET /audit/{id}

GET /audit/search

GET /audit/report

GET /audit/export

17.14 Replay API Traceability

API Group:

API-REP

Replay APIs

Includes:

POST /replay

GET /replay/{id}

GET /replay/history

GET /replay/report

GET /replay/validation

17.15 Audit Component Traceability

Component:

COMP-AUD-001

Audit Component

Maps To:

Audit Service

Audit APIs

Audit Repository

Audit Validators

Governance Controls

17.16 Replay Component Traceability

Component:

COMP-REP-001

Replay Component

Maps To:

Replay Engine

Replay APIs

Replay Repository

Replay Validators

Governance Controls

17.17 Audit Control Traceability

Audit Controls:

AUD-CTRL-001

Creation Audit

AUD-CTRL-002

Modification Audit

AUD-CTRL-003

Deletion Audit

AUD-CTRL-004

Access Audit

AUD-CTRL-005

Governance Audit

AUD-CTRL-006

Release Audit

17.18 Replay Control Traceability

Replay Controls:

REP-CTRL-001

Decision Replay

REP-CTRL-002

State Replay

REP-CTRL-003

Historical Replay

REP-CTRL-004

Governance Replay

REP-CTRL-005

Release Replay

17.19 Decision Reconstruction Traceability

Reconstruction Controls:

RECON-001

Input Reconstruction

RECON-002

Retrieval Reconstruction

RECON-003

Ranking Reconstruction

RECON-004

Recommendation Reconstruction

RECON-005

Governance Reconstruction

17.20 Audit Governance Traceability

Governance Controls:

GOV-AUD-001

Audit Governance

GOV-AUD-002

Retention Governance

GOV-AUD-003

Compliance Governance

GOV-AUD-004

Audit Review Governance

17.21 Replay Governance Traceability

Governance Controls:

GOV-REP-001

Replay Governance

GOV-REP-002

Replay Validation

GOV-REP-003

Historical Validation

GOV-REP-004

Replay Review Governance

17.22 Audit Testing Traceability

Verification Artifacts:

TEST-AUD-UNIT

Unit Tests

TEST-AUD-INT

Integration Tests

TEST-AUD-API

API Tests

TEST-AUD-COMP

Compliance Tests

TEST-AUD-UAT

UAT Tests

17.23 Replay Testing Traceability

Verification Artifacts:

TEST-REP-UNIT

Unit Tests

TEST-REP-INT

Integration Tests

TEST-REP-API

API Tests

TEST-REP-VALIDATION

Replay Validation Tests

TEST-REP-UAT

UAT Tests

17.24 Audit Security Traceability

Security Controls:

SEC-AUD-001

Authentication

SEC-AUD-002

Authorization

SEC-AUD-003

Access Controls

SEC-AUD-004

Audit Protection

SEC-AUD-005

Integrity Protection

17.25 Replay Security Traceability

Security Controls:

SEC-REP-001

Authentication

SEC-REP-002

Authorization

SEC-REP-003

Replay Access Controls

SEC-REP-004

Snapshot Protection

SEC-REP-005

Integrity Protection

17.26 Audit Reporting Traceability

Reports Supported:

Audit Summary Report

Compliance Audit Report

Governance Audit Report

Access Audit Report

Release Audit Report

17.27 Replay Reporting Traceability

Reports Supported:

Replay Summary Report

Decision Replay Report

Historical Replay Report

Governance Replay Report

Validation Report

17.28 Audit & Replay Release Traceability

Release Controls:

REL-AUD-001

Audit Validation

REL-REP-001

Replay Validation

REL-AUD-002

Compliance Validation

REL-AUD-003

Release Authorization

17.29 Audit & Replay Coverage Requirements

Every audit and replay capability shall map to:

Requirement

Architecture

Design

Database

API

Testing

Governance

Audit

Replay

Release

17.30 Audit & Replay Gap Analysis

Gaps exist when:

Missing Requirement Mapping

Missing Audit Controls

Missing Replay Controls

Missing Test Coverage

Missing Governance Validation

Missing Release Validation

All gaps require remediation.

17.31 Audit & Replay Impact Analysis

Changes shall trigger review of:

Audit Controls

Replay Controls

Snapshots

Reports

APIs

Tests

Governance Controls

Release Plans

17.32 Coverage Metrics

Required Metrics:

Audit Coverage %

Replay Coverage %

Governance Coverage %

Compliance Coverage %

Release Coverage %

Target:

100%

17.33 Governance Rules

RTM-AUD-GOV-01

Every audit capability must be traceable.

RTM-AUD-GOV-02

Every replay capability must be traceable.

RTM-AUD-GOV-03

Every audit capability must be tested.

RTM-AUD-GOV-04

Every replay capability must be tested.

RTM-AUD-GOV-05

Coverage must be measurable.

RTM-AUD-GOV-06

Release validation is mandatory.

17.34 Architectural Constraints

AC-RTM-AUD-01

Audit features without requirements are prohibited.

AC-RTM-AUD-02

Replay features without requirements are prohibited.

AC-RTM-AUD-03

Audit evidence cannot be deleted.

AC-RTM-AUD-04

Snapshot integrity must be preserved.

AC-RTM-AUD-05

Coverage gaps require remediation.

AC-RTM-AUD-06

Release approval requires complete audit and replay traceability.

17.35 Section Summary

This section defines the Audit & Replay Traceability Matrix and establishes mappings between audit requirements, replay requirements, architecture components, technical designs, APIs, database

entities, verification artifacts, governance controls, compliance controls, reconstruction controls, and release approvals.

The Audit & Replay Traceability Matrix ensures that TODP decisions, actions, recommendations, governance reviews, and releases remain transparent, reconstructable, auditable, reproducible, testable, and enterprise-ready throughout their lifecycle.

RTM V1 Progress

🔗 Section 1 — Document Overview & Traceability Scope

🔗 Section 2 — Traceability Framework

🔗 Section 3 — Requirement Classification Model

🔗 Section 4 — BRD Traceability Matrix

🔗 Section 5 — SRS Traceability Matrix

🔗 Section 6 — Architecture Traceability Matrix

🔗 Section 7 — Component Traceability Matrix

🔗 Section 8 — Database Traceability Matrix

🔗 Section 9 — API Traceability Matrix

🔗 Section 10 — Opportunity Module Traceability

🔗 Section 11 — Authority Module Traceability

🔗 Section 12 — Evidence Module Traceability

🔗 Section 13 — Retrieval Module Traceability

🔗 Section 14 — Ranking Module Traceability

🔗 Section 15 — Recommendation Module Traceability

🔗 Section 16 — Governance Traceability

🔗 Section 17 — Audit & Replay Traceability

▣ Section 18 — Test Coverage Traceability

RTM V1 — Section 18

Test Coverage Traceability

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

18

Version:

V1

Status:

APPROVED

18.1 Purpose

This section defines the Test Coverage Traceability Matrix for TODP.

The Test Coverage Traceability Matrix establishes traceability between business requirements, system requirements, architecture components, implementation artifacts, test artifacts, governance controls, and release validations.

The objective is to ensure:

- Complete verification coverage
- Requirement validation
- Risk reduction
- Governance compliance
- Audit readiness
- Release confidence

18.2 Test Coverage Objectives

RTM-TEST-01

Verify requirement coverage.

RTM-TEST-02

Verify implementation coverage.

RTM-TEST-03

Verify risk coverage.

RTM-TEST-04

Verify governance coverage.

RTM-TEST-05

Verify audit readiness.

RTM-TEST-06

Verify release readiness.

18.3 Test Traceability Model

Business Requirement



System Requirement



Architecture Component



Implementation Artifact



Test Specification



Test Case



Execution Evidence



Governance Validation



Release Approval

18.4 Test Domains

The following test domains shall be traced:

Unit Testing

Integration Testing

API Testing

Database Testing

Security Testing

Performance Testing

Governance Testing

Replay Testing

User Acceptance Testing

Release Testing

18.5 Requirement-to-Test Mapping Framework

Every requirement shall map to:

Minimum One Test Suite

Minimum One Test Case

Minimum One Execution Record

Minimum One Validation Artifact

Minimum One Release Validation Record

18.6 Opportunity Test Coverage Traceability

Requirement:

SRS-FR-OPP-001

Opportunity CRUD

Mapped Tests:

TEST-OPP-UNIT

Unit Tests



TEST-OPP-INT

Integration Tests



TEST-OPP-API

API Tests



TEST-OPP-UAT

UAT Tests



EXEC-OPP

Execution Evidence

18.7 Authority Test Coverage Traceability

Requirement:

SRS-FR-AUTH-001

Authority CRUD

Mapped Tests:

TEST-AUTH-UNIT

Unit Tests

↓

TEST-AUTH-INT

Integration Tests

↓

TEST-AUTH-API

API Tests



TEST-AUTH-UAT

UAT Tests



EXEC-AUTH

Execution Evidence

18.8 Evidence Test Coverage Traceability

Requirement:

SRS-FR-EVD-001

Evidence CRUD

Mapped Tests:

TEST-EVD-UNIT

Unit Tests



TEST-EVD-INT

Integration Tests



TEST-EVD-API

API Tests



TEST-EVD-UAT

UAT Tests



EXEC-EVD

Execution Evidence

18.9 Retrieval Test Coverage Traceability

Requirement:

SRS-FR-RET-001

Hybrid Retrieval

Mapped Tests:

TEST-RET-UNIT

Unit Tests

↓

TEST-RET-INT

Integration Tests

↓

TEST-RET-API

API Tests



TEST-RET-PERF

Performance Tests



EXEC-RET

Execution Evidence

18.10 Ranking Test Coverage Traceability

Requirement:

SRS-FR-RANK-001

Ranking Execution

Mapped Tests:

TEST-RANK-UNIT

Unit Tests



TEST-RANK-INT

Integration Tests



TEST-RANK-API

API Tests



TEST-RANK-PERF

Performance Tests



EXEC-RANK

Execution Evidence

18.11 Recommendation Test Coverage Traceability

Requirement:

SRS-FR-REC-001

Recommendation Generation

Mapped Tests:

TEST-REC-UNIT

Unit Tests

↓

TEST-REC-INT

Integration Tests



TEST-REC-API

API Tests



TEST-REC-UAT

UAT Tests



EXEC-REC

Execution Evidence

18.12 Governance Test Coverage Traceability

Requirement:

SRS-FR-GOV-001

Policy Enforcement

Mapped Tests:

TEST-GOV-UNIT

Unit Tests

↓

TEST-GOV-INT

Integration Tests

↓

TEST-GOV-COMP

Compliance Tests

↓

TEST-GOV-UAT

UAT Tests



EXEC-GOV

Execution Evidence

18.13 Audit Test Coverage Traceability

Requirement:

SRS-FR-AUD-001

Audit Logging

Mapped Tests:

TEST-AUD-UNIT

Unit Tests



TEST-AUD-INT

Integration Tests

↓

TEST-AUD-API

API Tests

↓

TEST-AUD-COMP

Compliance Tests

↓

EXEC-AUD

Execution Evidence

18.14 Replay Test Coverage Traceability

Requirement:

SRS-FR-REP-001

Replay Execution

Mapped Tests:

TEST-REP-UNIT

Unit Tests

↓

TEST-REP-INT

Integration Tests

↓

TEST-REP-VALIDATION

Replay Validation Tests

↓

TEST-REP-UAT

UAT Tests

↓

EXEC-REP

Execution Evidence

18.15 Security Test Coverage Traceability

Security Requirements:

SRS-NFR-SEC

Mapped Tests:

TEST-SEC-AUTH

Authentication Tests

TEST-SEC-RBAC

Authorization Tests

TEST-SEC-VULN

Vulnerability Tests

TEST-SEC-PEN

Penetration Tests

TEST-SEC-COMP

Security Compliance Tests

18.16 Database Test Coverage Traceability

Database Requirements



TEST-DB-SCHEMA

Schema Tests



TEST-DB-CONSTRAINT

Constraint Tests



TEST-DB-INTEGRITY

Integrity Tests



TEST-DB-BACKUP

Backup Tests



EXEC-DB

Execution Evidence

18.17 API Test Coverage Traceability

API Requirements



TEST-API-FUNCTIONAL

Functional Tests



TEST-API-CONTRACT

Contract Tests



TEST-API-NEGATIVE

Negative Tests



TEST-API-LOAD

Load Tests



EXEC-API

Execution Evidence

18.18 Performance Test Coverage Traceability

Performance Requirements



TEST-PERF-LOAD

Load Tests



TEST-PERF-STRESS

Stress Tests



TEST-PERF-SPIKE

Spike Tests



TEST-PERF-ENDURANCE

Endurance Tests



EXEC-PERF

Execution Evidence

18.19 Governance Validation Traceability

Every test execution shall map to:

Requirement



Test Case



Execution Record



Governance Review



Release Approval

18.20 Test Evidence Traceability

Execution evidence shall contain:

Execution ID

Test Case ID

Requirement ID

Execution Date

Execution Result

Approver

Evidence Location

18.21 Test Defect Traceability

Defects shall trace to:

Requirement



Test Case



Defect Record



Fix Record



Retest Evidence



Approval Record

18.22 Test Coverage Classification

Coverage Types:

Functional Coverage

Technical Coverage

Security Coverage

Performance Coverage

Governance Coverage

Compliance Coverage

Release Coverage

18.23 Coverage Calculation Rules

Coverage Formula:

Covered Requirements

÷

Total Requirements

×

100

Coverage Target:

100%

18.24 Coverage Thresholds

Critical Requirements

100%

High Requirements

100%

Medium Requirements

95%+

Low Requirements

90%+

18.25 Test Gap Analysis

Coverage gaps exist when:

Requirement Without Tests

Test Without Requirement

Missing Execution Evidence

Failed Governance Validation

Missing Release Validation

All gaps require remediation.

18.26 Regression Test Traceability

Regression Controls:

REG-001

Functional Regression

REG-002

API Regression

REG-003

Database Regression

REG-004

Security Regression

REG-005

Governance Regression

18.27 Release Validation Traceability

Release approval requires:

Successful Testing



Successful Governance Validation



Successful Compliance Validation



Successful Audit Validation



Release Authorization

18.28 Coverage Metrics

Required Metrics:

Requirement Coverage %

Test Coverage %

Defect Closure %

Governance Validation %

Compliance Validation %

Release Readiness %

Target:

100%

18.29 Governance Rules

RTM-TEST-GOV-01

Every requirement must be tested.

RTM-TEST-GOV-02

Every test must map to requirements.

RTM-TEST-GOV-03

Every execution must produce evidence.

RTM-TEST-GOV-04

Coverage must be measurable.

RTM-TEST-GOV-05

Critical requirements require 100% coverage.

RTM-TEST-GOV-06

Release approval requires test validation.

18.30 Architectural Constraints

AC-RTM-TEST-01

Untested requirements are prohibited.

AC-RTM-TEST-02

Requirements without evidence are prohibited.

AC-RTM-TEST-03

Coverage gaps require remediation.

AC-RTM-TEST-04

Failed critical tests block releases.

AC-RTM-TEST-05

Test evidence cannot be deleted.

AC-RTM-TEST-06

Release approval requires complete test traceability.

18.31 Section Summary

This section defines the Test Coverage Traceability Matrix and establishes mappings between requirements, architecture components, implementation artifacts, test cases, execution evidence, governance reviews, compliance validations, and release approvals.

The Test Coverage Traceability Matrix ensures that TODP requirements are fully verified, validated, governed, auditable, measurable, and release-ready throughout the software lifecycle.

RTM V1 Progress

▢ Section 1 — Document Overview & Traceability Scope

▢ Section 2 — Traceability Framework

▢ Section 3 — Requirement Classification Model

▢ Section 4 — BRD Traceability Matrix

▢ Section 5 — SRS Traceability Matrix

▢ Section 6 — Architecture Traceability Matrix

▢ Section 7 — Component Traceability Matrix

▢ Section 8 — Database Traceability Matrix

▢ Section 9 — API Traceability Matrix

▢ Section 10 — Opportunity Module Traceability

▢ Section 11 — Authority Module Traceability

▢ Section 12 — Evidence Module Traceability

▢ Section 13 — Retrieval Module Traceability

▢ Section 14 — Ranking Module Traceability

▢ Section 15 — Recommendation Module Traceability

▢ Section 16 — Governance Traceability

☐ Section 17 — Audit & Replay Traceability

☐ Section 18 — Test Coverage Traceability

☐ Section 19 — Release Readiness Traceability

RTM V1 — Section 19

Release Readiness Traceability

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

19

Version:

V1

Status:

APPROVED

19.1 Purpose

This section defines the Release Readiness Traceability Matrix for TODP.

Release Readiness Traceability ensures that every business requirement, system requirement, architectural component, implementation artifact, governance control, audit control, test artifact, and operational dependency has been validated before production deployment.

The objective is to ensure:

- Deployment confidence
- Operational readiness
- Governance approval
- Compliance validation
- Audit readiness
- Production stability

19.2 Release Readiness Objectives

RTM-REL-01

Verify release completeness.

RTM-REL-02

Verify deployment readiness.

RTM-REL-03

Verify governance approval.

RTM-REL-04

Verify compliance readiness.

RTM-REL-05

Verify audit readiness.

RTM-REL-06

Verify production readiness.

19.3 Release Traceability Model

Business Requirement



System Requirement



Implementation Artifact



Test Validation



Governance Validation



Compliance Validation



Audit Validation



Release Approval



Production Deployment

19.4 Release Domains

The following domains shall be traced:

Business Readiness

Technical Readiness

Security Readiness

Governance Readiness

Compliance Readiness

Audit Readiness

Operational Readiness

Production Readiness

19.5 Release Business Requirements

Primary Business Requirements:

BRD-REL-001

Support Production Releases

BRD-REL-002

Support Controlled Deployments

BRD-REL-003

Support Governance Approval

BRD-REL-004

Support Compliance Validation

BRD-REL-005

Support Operational Readiness

19.6 Release Functional Requirements

Mapped Requirements:

SRS-FR-REL-001

Release Validation

SRS-FR-REL-002

Deployment Approval

SRS-FR-REL-003

Rollback Support

SRS-FR-REL-004

Release Monitoring

SRS-FR-REL-005

Production Verification

19.7 Release Architecture Traceability

Architecture Components:

SAD-INFRA

Infrastructure Layer

SAD-OBS-001

Observability Layer

SAD-COMP-GOV

Governance Service

Responsibilities:

Release Support

Operational Validation

Monitoring

Governance Approval

Deployment Oversight

19.8 Release Design Traceability

Technical Design:

TDD-REL

Release Management Design

Includes:

Release Workflows

Approval Workflows

Rollback Procedures

Validation Procedures

Deployment Controls

19.9 Release Database Traceability

Database Entities:

DB-GOVERNANCE

Release Approvals

DB-AUDIT

Release Audit Records

DB-CONFIG

Deployment Configuration

Stores:

Release Metadata

Approvals

Validation Results

Audit References

Deployment References

19.10 Release API Traceability

API Group:

API-REL

Release Management APIs

Includes:

POST /releases

POST /releases/approve

POST /releases/deploy

POST /releases/rollback

GET /releases/history

19.11 Release Component Traceability

Component:

COMP-REL-001

Release Management Component

Maps To:

Release Services

Approval Services

Deployment Services

Rollback Services

Monitoring Services

19.12 Requirement Validation Traceability

Validation Controls:

REL-VAL-001

Requirement Coverage Validation

REL-VAL-002

Implementation Validation

REL-VAL-003

Verification Validation

REL-VAL-004

Traceability Validation

19.13 Test Validation Traceability

Validation Controls:

REL-TEST-001

Functional Validation

REL-TEST-002

Integration Validation

REL-TEST-003

Security Validation

REL-TEST-004

Performance Validation

REL-TEST-005

UAT Validation

19.14 Governance Validation Traceability

Validation Controls:

REL-GOV-001

Governance Review

REL-GOV-002

Approval Validation

REL-GOV-003

Policy Validation

REL-GOV-004

Exception Validation

19.15 Compliance Validation Traceability

Validation Controls:

REL-COMP-001

Compliance Review

REL-COMP-002

Control Validation

REL-COMP-003

Evidence Validation

REL-COMP-004

Compliance Approval

19.16 Audit Validation Traceability

Validation Controls:

REL-AUD-001

Audit Review

REL-AUD-002

Evidence Review

REL-AUD-003

Traceability Review

REL-AUD-004

Audit Approval

19.17 Security Validation Traceability

Validation Controls:

REL-SEC-001

Authentication Validation

REL-SEC-002

Authorization Validation

REL-SEC-003

Vulnerability Validation

REL-SEC-004

Penetration Validation

19.18 Infrastructure Validation Traceability

Validation Controls:

REL-INFRA-001

Database Validation

REL-INFRA-002

Cache Validation

REL-INFRA-003

Search Infrastructure Validation

REL-INFRA-004

Monitoring Validation

19.19 Deployment Traceability

Deployment Controls:

DEP-001

Deployment Approval

DEP-002

Deployment Execution

DEP-003

Deployment Verification

DEP-004

Deployment Monitoring

19.20 Rollback Traceability

Rollback Controls:

RB-001

Rollback Readiness

RB-002

Rollback Execution

RB-003

Rollback Validation

RB-004

Rollback Approval

19.21 Production Verification Traceability

Production Controls:

PROD-001

Smoke Validation

PROD-002

Health Validation

PROD-003

Monitoring Validation

PROD-004

Operational Validation

19.22 Release Evidence Traceability

Release evidence shall contain:

Release ID

Version

Approval Records

Validation Records

Audit Records

Deployment Records

Production Records

19.23 Release Reporting Traceability

Reports Supported:

Release Summary Report

Deployment Report

Validation Report

Compliance Report

Audit Report

Production Readiness Report

19.24 Release Approval Traceability

Approval Authorities:

Product Owner

Engineering Lead

QA Lead

Governance Lead

Security Lead

Release Manager

19.25 Release Coverage Requirements

Every release shall map to:

Requirements

Architecture

Implementation

Testing

Governance

Compliance

Audit

Operations

Production Validation

19.26 Release Gap Analysis

Release gaps exist when:

Missing Validation

Missing Approval

Missing Evidence

Missing Compliance Review

Missing Audit Review

Missing Production Verification

All gaps require remediation.

19.27 Release Impact Analysis

Changes to release requirements shall trigger review of:

Approvals

Validation Controls

Compliance Controls

Audit Controls

Deployment Plans

Rollback Plans

Monitoring Plans

19.28 Coverage Metrics

Required Metrics:

Release Coverage %

Approval Coverage %

Validation Coverage %

Compliance Coverage %

Audit Coverage %

Production Readiness %

Target:

100%

19.29 Governance Rules

RTM-REL-GOV-01

Every release must be traceable.

RTM-REL-GOV-02

Every release must be validated.

RTM-REL-GOV-03

Every release must be approved.

RTM-REL-GOV-04

Every release must be auditable.

RTM-REL-GOV-05

Coverage must be measurable.

RTM-REL-GOV-06

Production deployment requires approval.

19.30 Architectural Constraints

AC-RTM-REL-01

Releases without validation are prohibited.

AC-RTM-REL-02

Releases without approvals are prohibited.

AC-RTM-REL-03

Releases without audit evidence are prohibited.

AC-RTM-REL-04

Coverage gaps require remediation.

AC-RTM-REL-05

Release evidence cannot be deleted.

AC-RTM-REL-06

Production deployment requires complete release traceability.

19.31 Section Summary

This section defines the Release Readiness Traceability Matrix and establishes mappings between requirements, implementation artifacts, validation activities, governance reviews, compliance controls, audit evidence, approvals, deployments, rollbacks, and production verifications.

The Release Readiness Traceability Matrix ensures that TODP releases are fully validated, governed, compliant, auditable, operationally ready, and production-ready before deployment.

RTM V1 Progress

- 🔗 Section 1 — Document Overview & Traceability Scope
- 🔗 Section 2 — Traceability Framework
- 🔗 Section 3 — Requirement Classification Model
- 🔗 Section 4 — BRD Traceability Matrix
- 🔗 Section 5 — SRS Traceability Matrix
- 🔗 Section 6 — Architecture Traceability Matrix
- 🔗 Section 7 — Component Traceability Matrix
- 🔗 Section 8 — Database Traceability Matrix
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- 🔗 Section 11 — Authority Module Traceability
- 🔗 Section 12 — Evidence Module Traceability
- 🔗 Section 13 — Retrieval Module Traceability
- 🔗 Section 14 — Ranking Module Traceability
- 🔗 Section 15 — Recommendation Module Traceability
- 🔗 Section 16 — Governance Traceability
- 🔗 Section 17 — Audit & Replay Traceability
- 🔗 Section 18 — Test Coverage Traceability
- 🔗 Section 19 — Release Readiness Traceability
- 🔗 Section 20 — Governance & Compliance Traceability

RTM V1 — Section 20

Governance & Compliance Traceability

Document:

Requirements Traceability Matrix (RTM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

20

Version:

V1

Status:

APPROVED

20.1 Purpose

This section defines the Governance & Compliance Traceability Matrix for TODP.

The Governance & Compliance Traceability Matrix serves as the final enterprise traceability layer, ensuring that all business requirements, technical requirements, architectural decisions, implementations, testing activities, operational controls, governance controls, audit controls, and release activities are aligned with enterprise governance obligations and compliance expectations.

The objective is to ensure:

- Enterprise governance readiness
- Compliance readiness
- Audit readiness
- Risk accountability
- Control effectiveness
- Regulatory preparedness

20.2 Governance & Compliance Objectives

RTM-GC-01

Verify governance coverage.

RTM-GC-02

Verify compliance coverage.

RTM-GC-03

Verify control traceability.

RTM-GC-04

Verify audit readiness.

RTM-GC-05

Verify risk traceability.

RTM-GC-06

Verify enterprise readiness.

20.3 Governance & Compliance Traceability Model

Business Requirement

↓

System Requirement

↓

Control Requirement

↓

Governance Control

↓

Compliance Control

↓

Audit Evidence

↓

Risk Validation

↓

Release Authorization

20.4 Governance Domains

The following governance domains shall be traced:

Business Governance

Technology Governance

Security Governance

Data Governance

Operational Governance

Compliance Governance

Audit Governance

Release Governance

20.5 Compliance Domains

The following compliance domains shall be traced:

Access Control

Data Protection

Auditability

Retention

Change Management

Incident Management

Operational Monitoring

Release Management

20.6 Governance Business Requirements

Primary Business Requirements:

BRD-GOV-001

Govern Platform Operations

BRD-GOV-002

Enforce Enterprise Policies

BRD-GOV-003

Support Risk Management

BRD-GOV-004

Support Audit Readiness

BRD-GOV-005

Support Compliance Readiness

20.7 Compliance Functional Requirements

Mapped Requirements:

SRS-FR-COMP-001

Compliance Validation

SRS-FR-COMP-002

Control Enforcement

SRS-FR-COMP-003

Evidence Collection

SRS-FR-COMP-004

Compliance Reporting

SRS-FR-COMP-005

Compliance Review

20.8 Governance Architecture Traceability

Architecture Components:

SAD-COMP-GOV

Governance Service

SAD-COMP-AUD

Audit Service

SAD-OBS-001

Observability Layer

Responsibilities:

Policy Enforcement

Control Validation

Audit Collection

Compliance Reporting

Risk Monitoring

20.9 Governance Design Traceability

Technical Design:

TDD-GOV

Governance Design

Includes:

Control Models

Policy Models

Risk Models

Compliance Models

Audit Models

20.10 Governance Database Traceability

Database Entities:

DB-GOVERNANCE

Governance Records

DB-AUDIT

Audit Records

DB-SNAPSHOT

Replay Records

Stores:

Control Evidence

Approval Evidence

Compliance Evidence

Audit Evidence

Risk Evidence

20.11 Governance API Traceability

API Group:

API-GOV

Governance APIs

Includes:

/governance

/compliance

/controls

/reviews

/reports

20.12 Policy Traceability

Policy Controls:

POL-001

Business Policies

POL-002

Security Policies

POL-003

Data Policies

POL-004

Operational Policies

POL-005

Release Policies

20.13 Control Traceability

Control Categories:

CTRL-001

Preventive Controls

CTRL-002

Detective Controls

CTRL-003

Corrective Controls

CTRL-004

Compensating Controls

20.14 Access Control Traceability

Control Domain:

Access Governance

Controls:

Authentication Controls

Authorization Controls

Role Controls

Privilege Controls

Access Review Controls

20.15 Data Governance Traceability

Control Domain:

Data Governance

Controls:

Ownership Controls

Classification Controls

Retention Controls

Deletion Controls

Protection Controls

20.16 Change Management Traceability

Control Domain:

Change Governance

Controls:

Requirement Changes

Design Changes

Code Changes

Database Changes

Release Changes

20.17 Incident Management Traceability

Control Domain:

Operational Governance

Controls:

Incident Detection

Incident Classification

Incident Response

Incident Recovery

Incident Review

20.18 Risk Management Traceability

Risk Controls:

RISK-001

Business Risks

RISK-002

Technical Risks

RISK-003

Security Risks

RISK-004

Compliance Risks

RISK-005

Operational Risks

20.19 Audit Traceability

Audit Controls:

AUD-GC-001

Control Audits

AUD-GC-002

Governance Audits

AUD-GC-003

Compliance Audits

AUD-GC-004

Security Audits

AUD-GC-005

Release Audits

20.20 Evidence Traceability

Evidence Types:

Approval Evidence

Control Evidence

Testing Evidence

Compliance Evidence

Audit Evidence

Release Evidence

20.21 Compliance Reporting Traceability

Reports Supported:

Governance Report

Compliance Report

Risk Report

Audit Report

Control Effectiveness Report

Release Readiness Report

20.22 Governance Testing Traceability

Verification Artifacts:

TEST-GOV-COMP

Compliance Tests

TEST-GOV-AUD

Audit Tests

TEST-GOV-RISK

Risk Validation Tests

TEST-GOV-CTRL

Control Validation Tests

20.23 Compliance Validation Traceability

Validation Activities:

Control Validation

Evidence Validation

Policy Validation

Audit Validation

Release Validation

20.24 Governance Approval Traceability

Approval Authorities:

Product Owner

Engineering Lead

Governance Lead

Security Lead

Compliance Lead

Release Manager

20.25 Enterprise Framework Alignment

TODP traceability shall support readiness for:

SOC2-style controls

ISO27001-style controls

Enterprise Procurement Reviews

Customer Security Reviews

Internal Governance Reviews

20.26 Coverage Requirements

Every governance and compliance control shall map to:

Requirement

Control

Evidence

Validation

Audit

Risk Assessment

Release Authorization

20.27 Governance Gap Analysis

Governance gaps exist when:

Missing Controls

Missing Evidence

Missing Validation

Missing Audits

Missing Risk Reviews

Missing Release Authorization

All gaps require remediation.

20.28 Governance Impact Analysis

Changes shall trigger review of:

Policies

Controls

Risk Registers

Compliance Reviews

Audit Reviews

Release Approvals

20.29 Coverage Metrics

Required Metrics:

Governance Coverage %

Compliance Coverage %

Audit Coverage %

Risk Coverage %

Control Coverage %

Release Authorization Coverage %

Target:

100%

20.30 Governance Rules

RTM-GC-GOV-01

Every control must be traceable.

RTM-GC-GOV-02

Every control must have evidence.

RTM-GC-GOV-03

Every control must be validated.

RTM-GC-GOV-04

Every control must be auditable.

RTM-GC-GOV-05

Coverage must be measurable.

RTM-GC-GOV-06

Enterprise release authorization is mandatory.

20.31 Architectural Constraints

AC-RTM-GC-01

Controls without requirements are prohibited.

AC-RTM-GC-02

Controls without evidence are prohibited.

AC-RTM-GC-03

Controls without validation are prohibited.

AC-RTM-GC-04

Coverage gaps require remediation.

AC-RTM-GC-05

Governance evidence cannot be deleted.

AC-RTM-GC-06

Enterprise release approval requires complete governance traceability.

20.32 RTM Completion Statement

RTM V1 provides complete end-to-end traceability across:

Business Requirements

System Requirements

Architecture Components

Technical Components

Database Entities

APIs

Testing Artifacts

Governance Controls

Audit Controls

Replay Controls

Release Controls

Compliance Controls

Enterprise Controls

The RTM establishes full forward and backward traceability from business objectives through implementation, verification, governance validation, audit validation, compliance validation, and production release.

20.33 RTM Success Criteria

RTM Success requires:

100% Requirement Coverage

100% Architecture Coverage

100% Implementation Coverage

100% Test Coverage

100% Governance Coverage

100% Audit Coverage

100% Release Coverage

100% Compliance Coverage

20.34 Final Section Summary

This section defines the Governance & Compliance Traceability Matrix and establishes the final enterprise traceability layer for TODP.

The Governance & Compliance Traceability Matrix ensures that all platform capabilities, controls, validations, approvals, audits, and releases remain traceable, governable, compliant, auditable, measurable, and enterprise-ready throughout the entire system lifecycle.

RTM V1 Progress

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🔗 Section 13 — Retrieval Module Traceability

🔗 Section 14 — Ranking Module Traceability

🔗 Section 15 — Recommendation Module Traceability

🔗 Section 16 — Governance Traceability

☐ Section 17 — Audit & Replay Traceability

☐ Section 18 — Test Coverage Traceability

☐ Section 19 — Release Readiness Traceability

☐ Section 20 — Governance & Compliance Traceability

☐ RTM V1 COMPLETE